

Notes on the Modern Theory of Linear Models

Notation and conventions. Throughout, $\|\mathbf{v}\| = \|\mathbf{v}\|_2 = (\sum_i v_i^2)^{1/2}$ is the Euclidean norm, $\|\mathbf{A}\|_{\text{op}} = \sup_{\|\mathbf{v}\|=1} \|\mathbf{A}\mathbf{v}\|$ is the operator (spectral) norm, $\|\mathbf{A}\|_{\text{F}} = (\text{tr}(\mathbf{A}'\mathbf{A}))^{1/2}$ is the Frobenius norm, and $\sigma_{\min}(\mathbf{A})$, $\sigma_{\max}(\mathbf{A})$ denote the smallest and largest singular values of $\mathbf{A}$. For a symmetric matrix $\mathbf{A}$, $\lambda_{\min}(\mathbf{A})$ and $\lambda_{\max}(\mathbf{A})$ denote its extreme eigenvalues; then $\|\mathbf{A}\|_{\text{op}} = \max(|\lambda_{\min}(\mathbf{A})|, |\lambda_{\max}(\mathbf{A})|)$. We write $a \lesssim b$ to mean $a \leq Cb$ for a universal constant $C > 0$. For two symmetric matrices, $\mathbf{A} \preceq \mathbf{A}'$ means $\mathbf{A}' - \mathbf{A}$ is positive semi-definite. We write $a \wedge b = \min(a, b)$ and $a \vee b = \max(a, b)$. The standard Gaussian random vector in $\mathbb{R}^n$ is denoted $\mathbf{g} \sim N(\mathbf{0}, \mathbf{I}_n)$. In Part 3, p replaces $k + 1$ for the total number of regressors (including the intercept, if present).

Prerequisites. This document assumes familiarity with the material in *Notes on the Classical Theory of Linear Models*. The primary references are R. Vershynin, *High-Dimensional Probability* (HDP, 2018), and M. J. Wainwright, *High-Dimensional Statistics* (HDS, 2019).

Part I

The Toolkit: Sub-Gaussian and Sub-Exponential Variables

1 Sub-Gaussian Random Variables

1.1 Motivation: Why Go Beyond Normality?

Remark. In the classical linear model (Assumption (A5)), exact inference rests on the normality of $\boldsymbol{\varepsilon}$. The Gauss–Markov result (Theorem 2.6 in the classical notes) needs only (A1)–(A4), but the pivotal distributions t_{n-k-1} and $F_{k, n-k-1}$ demand Gaussian errors. The modern programme asks: *what minimal distributional property replaces normality to yield sharp, non-asymptotic concentration?* The answer is the sub-Gaussian condition: the tails of the random variable decay at least as fast as those of a Gaussian. This class includes all bounded random variables, all Gaussian random variables, and many other distributions encountered in practice.

1.2 Definition and Equivalent Characterizations

Definition 1.1 (Sub-Gaussian Random Variable). A real-valued random variable X with $\mathbb{E}[X] = 0$ is called **sub-Gaussian** if there exists a constant $K > 0$ such that

$$\mathbb{E}[e^{tX}] \leq e^{K^2 t^2 / 2} \quad \text{for all } t \in \mathbb{R}. \quad (1)$$

The **sub-Gaussian norm** (or ψ_2 -norm) of X is

$$\|X\|_{\psi_2} = \inf \left\{ t > 0 : \mathbb{E}[e^{X^2/t^2}] \leq 2 \right\}. \quad (2)$$

Theorem 1.1 (Equivalent Characterizations of Sub-Gaussianity). Let X be a random variable with $\mathbb{E}[X] = 0$. The following properties are equivalent, in the sense that if one holds with constant K_i , then each of the others holds with a constant $K_j = C_{ij}K_i$ for a universal factor $C_{ij} > 0$:

- (i) **MGF bound:** $\mathbb{E}[e^{tX}] \leq e^{K_1^2 t^2 / 2}$ for all $t \in \mathbb{R}$.
- (ii) **Tail bound:** $\mathbb{P}(|X| \geq t) \leq 2e^{-t^2/(2K_2^2)}$ for all $t \geq 0$.
- (iii) **Moment bound:** $(\mathbb{E}[|X|^p])^{1/p} \leq K_3 \sqrt{p}$ for all $p \geq 1$.

(iv) ψ_2 -**norm finite**: $\|X\|_{\psi_2} \leq K_4$.

Moreover, $\|X\|_{\psi_2}$ is equivalent (up to universal constants) to each of the optimal K_i .

Proof. We prove the cycle (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii). By Markov's inequality and (i), for any $t > 0$ and $\lambda > 0$:

$$\mathbf{P}(X \geq t) = \mathbf{P}(e^{\lambda X} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbf{E}[e^{\lambda X}] \leq e^{-\lambda t + K_1^2 \lambda^2 / 2}.$$

Optimizing over λ : set $\frac{d}{d\lambda}(-\lambda t + K_1^2 \lambda^2 / 2) = -t + K_1^2 \lambda = 0$, giving $\lambda^* = t/K_1^2$. Substituting:

$$\mathbf{P}(X \geq t) \leq e^{-t^2/(2K_1^2)}.$$

Applying the same argument to $-X$ (which also satisfies (i) with the same constant, since $\mathbf{E}[e^{-tX}] = \mathbf{E}[e^{t(-X)}]$ and $(-X)$ has mean zero):

$$\mathbf{P}(|X| \geq t) \leq 2e^{-t^2/(2K_1^2)}.$$

So $K_2 = K_1$ works.

(ii) $\Rightarrow$ (iii). For any $p \geq 1$, the layer-cake formula gives:

$$\mathbf{E}[|X|^p] = \int_0^\infty p t^{p-1} \mathbf{P}(|X| \geq t) dt \leq \int_0^\infty p t^{p-1} \cdot 2e^{-t^2/(2K_2^2)} dt. \quad (3)$$

Substituting $s = t^2/(2K_2^2)$, so $t = K_2\sqrt{2s}$ and $dt = K_2 ds/\sqrt{2s}$:

$$\mathbf{E}[|X|^p] \leq 2p \int_0^\infty (K_2\sqrt{2s})^{p-1} e^{-s} \frac{K_2}{\sqrt{2s}} ds = p(K_2\sqrt{2})^p \Gamma(p/2).$$

For $p \geq 2$ we use the bound $\Gamma(\alpha) \leq 2\alpha^{\alpha-1}$, valid for $\alpha \geq 1$. To see it, write $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$ and bound $t^{\alpha-1} e^{-t/2}$ by its maximum over $t \geq 0$, attained at $t = 2(\alpha-1)$ with value $(2(\alpha-1)/e)^{\alpha-1}$; then

$$\Gamma(\alpha) = \int_0^\infty (t^{\alpha-1} e^{-t/2}) e^{-t/2} dt \leq \left(\frac{2(\alpha-1)}{e} \right)^{\alpha-1} \int_0^\infty e^{-t/2} dt = 2 \left(\frac{2(\alpha-1)}{e} \right)^{\alpha-1} \leq 2\alpha^{\alpha-1},$$

the last step using $\alpha-1 \leq \alpha$ and $2/e < 1$. Thus $\Gamma(p/2) \leq 2(p/2)^{p/2-1}$, and substituting:

$$\mathbf{E}[|X|^p] \leq p(K_2\sqrt{2})^p \cdot 2(p/2)^{p/2-1} = 4(K_2^2)^{p/2} p^{p/2}.$$

Taking p -th roots: $(\mathbf{E}[|X|^p])^{1/p} \leq 4^{1/p} K_2 \sqrt{p} \leq 2K_2 \sqrt{p}$ for $p \geq 2$. For $p = 1$: $\mathbf{E}[|X|] \leq 2 \int_0^\infty e^{-t^2/(2K_2^2)} dt = K_2 \sqrt{2\pi} \leq 3K_2$. So $K_3 = 3K_2$ works uniformly for all $p \geq 1$.

(iii) $\Rightarrow$ (iv). From (iii), $\mathbf{E}[|X|^{2k}] \leq (K_3\sqrt{2k})^{2k} = K_3^{2k} (2k)^k$. Using the Taylor expansion:

$$\mathbf{E}[e^{X^2/t^2}] = \sum_{k=0}^\infty \frac{\mathbf{E}[|X|^{2k}]}{k! t^{2k}} \leq 1 + \sum_{k=1}^\infty \frac{K_3^{2k} (2k)^k}{k! t^{2k}}.$$

Since $k! \geq (k/e)^k$ (Stirling's lower bound), $k^k/k! \leq e^k$. Hence $(2k)^k/k! = 2^k \cdot k^k/k! \leq (2e)^k$, and:

$$\frac{K_3^{2k} (2k)^k}{k! t^{2k}} \leq \left(\frac{2eK_3^2}{t^2} \right)^k.$$

Choosing $t^2 = 4eK_3^2$: each term is $\leq 2^{-k}$, so $\mathbf{E}[e^{X^2/t^2}] \leq 1 + 1 = 2$. Hence $\|X\|_{\psi_2} \leq 2\sqrt{e} K_3$.

(iv) $\Rightarrow$ (i). Let $K = \|X\|_{\psi_2}$, so $\mathbf{E}[e^{X^2/K^2}] \leq 2$.

Step 1: Obtain a tail bound. By Markov's inequality: for $u \geq 0$,

$$\mathbb{P}(|X| \geq u) = \mathbb{P}(e^{X^2/K^2} \geq e^{u^2/K^2}) \leq e^{-u^2/K^2} \mathbb{E}[e^{X^2/K^2}] \leq 2e^{-u^2/K^2}.$$

This is the tail bound (ii) with $K_2 = K/\sqrt{2}$.

Step 2: Obtain moment bounds. By the implication (ii) $\Rightarrow$ (iii) (already proved), $(\mathbb{E}[|X|^p])^{1/p} \leq CK\sqrt{p}$ for all $p \geq 1$, hence $|\mathbb{E}[X^p]| \leq \mathbb{E}[|X|^p] \leq (CK)^p p^{p/2}$.

Step 3: Bound the MGF via Taylor series. Since $\mathbb{E}[X] = 0$:

$$\mathbb{E}[e^{tX}] = 1 + \sum_{p=2}^{\infty} \frac{t^p \mathbb{E}[X^p]}{p!} \leq 1 + \sum_{p=2}^{\infty} \frac{|t|^p (CK)^p p^{p/2}}{p!}.$$

From Stirling's lower bound $p! \geq (p/e)^p$, we obtain $p^{p/2}/p! \leq e^p/p^{p/2}$, so

$$\frac{(CK|t|)^p p^{p/2}}{p!} \leq \frac{(CK|t|)^p e^p}{p^{p/2}} = \left(\frac{CeK|t|}{\sqrt{p}} \right)^p.$$

For $p \geq 2$ and $|t| \leq 1/(2CeK)$, write $r := CeK|t| \leq 1/2$; since $\sqrt{p} \geq 1$, each term is $\leq r^p$, and summing the geometric tail keeps the t^2 scaling:

$$\mathbb{E}[e^{tX}] \leq 1 + \sum_{p=2}^{\infty} r^p = 1 + \frac{r^2}{1-r} \leq 1 + 2r^2 \leq e^{2r^2} = e^{2C^2e^2K^2t^2}, \quad \text{for } |t| \leq \frac{1}{2CeK},$$

using $r/(1-r) \leq 2r$ (as $r \leq 1/2$) and $1+x \leq e^x$. This is already of the form $e^{(\text{const})K^2t^2}$ needed below; a crude constant bound $\mathbb{E}[e^{tX}] < 2$ would not capture the t^2 decay as $t \rightarrow 0$.

Step 4: Extend to all t . For $|t| > 1/(2CeK)$, we use the tail bound from Step 1. Split the MGF:

$$\mathbb{E}[e^{tX}] = \mathbb{E}[e^{tX} \mathbf{1}_{|X| \leq R}] + \mathbb{E}[e^{tX} \mathbf{1}_{|X| > R}]$$

with $R = K^2|t|$. For the first term: $e^{tX} \mathbf{1}_{|X| \leq R} \leq e^{|t|R} = e^{K^2t^2}$. For the second:

$$\begin{aligned} \mathbb{E}[e^{tX} \mathbf{1}_{|X| > R}] &\leq \mathbb{E}[e^{|t||X|} \mathbf{1}_{|X| > R}] = |t| \int_R^{\infty} e^{|t|u} \mathbb{P}(|X| \geq u) du + e^{|t|R} \mathbb{P}(|X| \geq R) \\ &\leq |t| \int_R^{\infty} e^{|t|u} \cdot 2e^{-u^2/K^2} du + 2e^{|t|R - R^2/K^2}. \end{aligned}$$

In the exponent: $|t|u - u^2/K^2 = -(u/K - K|t|/2)^2 + K^2t^2/4$. So all terms are bounded by $Ce^{K^2t^2/4}$. (polynomial in $|t|K$), giving $\mathbb{E}[e^{tX}] \leq e^{C'K^2t^2}$ for a universal C' .

Combining Steps 3–4: $\mathbb{E}[e^{tX}] \leq e^{C''K^2t^2/2}$ for all $t \in \mathbb{R}$, with $K_1 = \sqrt{2C''}K_4$. $\square$

Example 1.1 (Sub-Gaussian Random Variables).

- (1) **Gaussian.** If $X \sim N(0, \sigma^2)$, then $\mathbb{E}[e^{tX}] = e^{\sigma^2 t^2/2}$, so X is sub-Gaussian with $K_1 = \sigma$. Moreover, $\|X\|_{\psi_2} = C\sigma$ for a universal C .
- (2) **Bounded (Rademacher).** Let X take values ± 1 with equal probability. Then $\mathbb{E}[e^{tX}] = \cosh(t) \leq e^{t^2/2}$, so X is sub-Gaussian with $K_1 = 1$.
- (3) **Bounded (Hoeffding's lemma).** If $a \leq X \leq b$ and $\mathbb{E}[X] = 0$, then $\mathbb{E}[e^{tX}] \leq e^{(b-a)^2 t^2/8}$, so X is sub-Gaussian with $K_1 = (b-a)/2$. The proof is given in Lemma 2.1.
- (4) **Non-example (Exponential).** If $X \sim \text{Exp}(1) - 1$ (centered), then $\mathbb{E}[e^{tX}] = e^{-t}/(1-t)$ for $t < 1$, which grows faster than any e^{Kt^2} as $t \rightarrow 1^-$. Hence X is *not* sub-Gaussian; it is sub-exponential (Definition 4.1).

1.3 Properties of Sub-Gaussian Random Variables

Proposition 1.1 (Closure Properties).

- (i) If X is sub-Gaussian and $a \in \mathbb{R}$, then aX is sub-Gaussian with $\|aX\|_{\psi_2} = |a| \|X\|_{\psi_2}$.
- (ii) If $X_1, \dots, X_n$ are independent sub-Gaussian random variables with $\mathbb{E}[X_i] = 0$, then $S = \sum_{i=1}^n X_i$ is sub-Gaussian with $\|S\|_{\psi_2}^2 \leq C \sum_{i=1}^n \|X_i\|_{\psi_2}^2$.

Proof. (i). Immediate from $\mathbb{E}[e^{(aX)^2/t^2}] = \mathbb{E}[e^{X^2/(t/|a|)^2}]$, which is ≤ 2 when $t/|a| \geq \|X\|_{\psi_2}$, i.e., $t \geq |a| \|X\|_{\psi_2}$.

(ii). By independence:

$$\mathbb{E}[e^{tS}] = \prod_{i=1}^n \mathbb{E}[e^{tX_i}] \leq \prod_{i=1}^n e^{K_i^2 t^2/2} = \exp\left(\frac{t^2}{2} \sum_{i=1}^n K_i^2\right),$$

where $K_i = C\|X_i\|_{\psi_2}$ from the MGF characterization. Hence S is sub-Gaussian with $K^2 = \sum K_i^2$, and $\|S\|_{\psi_2}^2 \leq C' \sum \|X_i\|_{\psi_2}^2$. $\square$

1.4 Variance of Sub-Gaussian Variables

Proposition 1.2. If X is sub-Gaussian with $\mathbb{E}[X] = 0$, then $\text{var}(X) = \mathbb{E}[X^2] \leq C\|X\|_{\psi_2}^2$ for a universal constant C .

Proof. From the moment bound (Theorem 1.1(iii)) with $p = 2$: $(\mathbb{E}[X^2])^{1/2} \leq K_3\sqrt{2}$. Since $K_3 \leq C\|X\|_{\psi_2}$, squaring gives $\mathbb{E}[X^2] \leq 2C^2\|X\|_{\psi_2}^2$. $\square$

2 Hoeffding's Inequality

2.1 Hoeffding's Lemma

Lemma 2.1 (Hoeffding's Lemma). Let X be a random variable with $\mathbb{E}[X] = 0$ and $a \leq X \leq b$ almost surely. Then for all $t \in \mathbb{R}$:

$$\mathbb{E}[e^{tX}] \leq \exp\left(\frac{t^2(b-a)^2}{8}\right). \quad (4)$$

Proof. If $a = b$, then $\mathbb{E}[X] = 0$ forces $X = a = b = 0$ almost surely, and both sides of (4) equal 1; assume henceforth $a < b$.

Step 1: Convexity bound. Since e^{tx} is convex in x , for any $x \in [a, b]$:

$$e^{tx} \leq \frac{b-x}{b-a} e^{ta} + \frac{x-a}{b-a} e^{tb}.$$

Taking expectations and using $\mathbb{E}[X] = 0$:

$$\mathbb{E}[e^{tX}] \leq \frac{b}{b-a} e^{ta} + \frac{-a}{b-a} e^{tb} = \frac{b e^{ta} - a e^{tb}}{b-a}. \quad (5)$$

Step 2: Logarithmic bound. Let $h = t(b-a)$, $\theta = -a/(b-a) \in [0, 1]$. Then $1 - \theta = b/(b-a)$. Rewrite (5):

$$\mathbb{E}[e^{tX}] \leq (1 - \theta) e^{-\theta h} + \theta e^{(1-\theta)h} = e^{-\theta h} [(1 - \theta) + \theta e^h].$$

Define $\varphi(h) = -\theta h + \ln[(1 - \theta) + \theta e^h]$. Then $\mathbb{E}[e^{tX}] \leq e^{\varphi(h)}$.

Step 3: Taylor expansion of φ . We have $\varphi(0) = 0$ and $\varphi'(0) = -\theta + \theta = 0$. Computing the second derivative:

$$\varphi''(h) = \frac{\theta(1-\theta)e^h}{[(1-\theta) + \theta e^h]^2}.$$

Let $u = \theta e^h / [(1-\theta) + \theta e^h] \in (0, 1)$. Then $\varphi''(h) = u(1-u) \leq 1/4$ (since $u(1-u) \leq 1/4$ for $u \in [0, 1]$). By Taylor's theorem with remainder:

$$\varphi(h) = \varphi(0) + h\varphi'(0) + \frac{h^2}{2}\varphi''(\xi) \leq \frac{h^2}{8}$$

for some ξ between 0 and h . Hence $\mathbb{E}[e^{tX}] \leq e^{h^2/8} = e^{t^2(b-a)^2/8}$. $\square$

2.2 Hoeffding's Inequality

Theorem 2.1 (Hoeffding's Inequality). Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}[X_i] = 0$ and $a_i \leq X_i \leq b_i$ almost surely for each i . Then for all $t \geq 0$:

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right). \quad (6)$$

Proof. Step 1: One-sided bound via Chernoff. For $\lambda > 0$:

$$\begin{aligned} \mathbb{P}\left(\sum_{i=1}^n X_i \geq t\right) &= \mathbb{P}(e^{\lambda \sum X_i} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E}[e^{\lambda \sum X_i}] \\ &= e^{-\lambda t} \prod_{i=1}^n \mathbb{E}[e^{\lambda X_i}] \leq e^{-\lambda t} \prod_{i=1}^n e^{\lambda^2 (b_i - a_i)^2 / 8}, \end{aligned} \quad (7)$$

where the last inequality uses Hoeffding's Lemma (Lemma 2.1) applied to each X_i .

Step 2: Optimize λ . From (7):

$$\mathbb{P}\left(\sum X_i \geq t\right) \leq \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2\right).$$

Let $\sigma_H^2 = \frac{1}{4} \sum (b_i - a_i)^2$. Setting $\frac{d}{d\lambda}(-\lambda t + \lambda^2 \sigma_H^2 / 2) = -t + \lambda \sigma_H^2 = 0$ gives $\lambda^* = t / \sigma_H^2$:

$$\mathbb{P}\left(\sum X_i \geq t\right) \leq \exp\left(-\frac{t^2}{2\sigma_H^2}\right) = \exp\left(-\frac{2t^2}{\sum (b_i - a_i)^2}\right).$$

Step 3: Two-sided bound. Apply the same argument to $-X_1, \dots, -X_n$ to get $\mathbb{P}(\sum X_i \leq -t) \leq \exp(-2t^2 / \sum (b_i - a_i)^2)$. By the union bound, $\mathbb{P}(|\sum X_i| \geq t) \leq 2 \exp(-2t^2 / \sum (b_i - a_i)^2)$. $\square$

Corollary 2.1 (Hoeffding for i.i.d. Bounded Variables). Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $a \leq X_i \leq b$. Then for $\bar{X} = \frac{1}{n} \sum X_i$:

$$\mathbb{P}(|\bar{X} - \mu| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{(b-a)^2}\right). \quad (8)$$

Proof. Apply Theorem 2.1 to $Y_i = X_i - \mu$ (so $\mathbb{E}[Y_i] = 0$, $a - \mu \leq Y_i \leq b - \mu$, and $b_i - a_i = b - a$ for all i). Then $\sum Y_i = n(\bar{X} - \mu)$, and the bound gives $\mathbb{P}(|n(\bar{X} - \mu)| \geq nt) \leq 2 \exp(-2n^2 t^2 / (n(b-a)^2))$. $\square$

3 Sub-Gaussian Concentration for Sums and Quadratic Forms

3.1 General Sub-Gaussian Hoeffding Inequality

Theorem 3.1 (General Hoeffding Inequality for Sub-Gaussians). Let $X_1, \dots, X_n$ be independent, mean-zero, sub-Gaussian random variables. Then for all $t \geq 0$:

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left(-\frac{ct^2}{\sum_{i=1}^n \|X_i\|_{\psi_2}^2}\right), \quad (9)$$

where $c > 0$ is a universal constant.

Proof. By the MGF characterization (Theorem 1.1(i)), each X_i satisfies $\mathbb{E}[e^{\lambda X_i}] \leq e^{C^2 \|X_i\|_{\psi_2}^2 \lambda^2 / 2}$. The proof then follows the Chernoff-optimize pattern of Theorem 2.1, with $(b_i - a_i)^2 / 4$ replaced by $C^2 \|X_i\|_{\psi_2}^2$:

$$\mathbb{P}\left(\sum X_i \geq t\right) \leq \exp\left(-\lambda t + \frac{C^2 \lambda^2}{2} \sum \|X_i\|_{\psi_2}^2\right).$$

Setting $\lambda^* = t / (C^2 \sum \|X_i\|_{\psi_2}^2)$ gives $\mathbb{P}(\sum X_i \geq t) \leq \exp(-t^2 / (2C^2 \sum \|X_i\|_{\psi_2}^2))$, and the two-sided bound follows by the union bound with $c = 1/(2C^2)$. $\square$

3.2 Sub-Gaussian Random Vectors

Definition 3.1 (Sub-Gaussian Random Vector). A random vector $\mathbf{x} \in \mathbb{R}^p$ is **sub-Gaussian** if all one-dimensional marginals are sub-Gaussian. Formally, there exists $K > 0$ such that

$$\|\langle \mathbf{x}, \mathbf{v} \rangle\|_{\psi_2} \leq K \|\mathbf{v}\| \quad \text{for all } \mathbf{v} \in \mathbb{R}^p. \quad (10)$$

The smallest such K is the **sub-Gaussian norm** of $\mathbf{x}$, denoted $\|\mathbf{x}\|_{\psi_2}$.

Example 3.1.

- (1) If $\mathbf{x} \sim N(\mathbf{0}, \Sigma)$, then $\langle \mathbf{x}, \mathbf{v} \rangle \sim N(0, \mathbf{v}' \Sigma \mathbf{v})$ has $\|\langle \mathbf{x}, \mathbf{v} \rangle\|_{\psi_2} = C(\mathbf{v}' \Sigma \mathbf{v})^{1/2} \leq C \|\Sigma\|_{\text{op}}^{1/2} \|\mathbf{v}\|$. So $\mathbf{x}$ is sub-Gaussian with $\|\mathbf{x}\|_{\psi_2} \leq C \|\Sigma\|_{\text{op}}^{1/2}$.
- (2) If $\mathbf{x} = (\varepsilon_1, \dots, \varepsilon_p)'$ where ε_i are independent Rademacher, then $\langle \mathbf{x}, \mathbf{v} \rangle = \sum v_i \varepsilon_i$ has $\|\langle \mathbf{x}, \mathbf{v} \rangle\|_{\psi_2} \leq C \|\mathbf{v}\|$ by Proposition 1.1(ii).

3.3 Hanson–Wright Inequality for Quadratic Forms

In this subsection we prove the Hanson–Wright concentration inequality for quadratic forms $\mathbf{x}' \mathbf{A} \mathbf{x}$, where $\mathbf{x}$ has independent sub-Gaussian coordinates. The result drives nearly every application in Parts 3–4. We give a complete, self-contained proof, with no step deferred to an external reference.

The presentation has three parts. (i) The Gaussian case (Theorem 3.2) is proved by computing the moment generating function (MGF) of a weighted sum of independent χ_1^2 variables, using the spectral decomposition of $\mathbf{A}$. (ii) The sub-Gaussian case (Theorem 3.3) is proved in eight steps. The two substantive ingredients beyond the Gaussian argument — a *decoupling lemma* and an *MGF lemma* for sub-Gaussian quadratic forms — are stated and proved in full; the MGF lemma uses the Hubbard–Stratonovich identity to reduce the sub-Gaussian quadratic-form MGF directly to the Gaussian one. (iii) We apply the result.

Gaussian case (full proof)

Theorem 3.2 (Hanson–Wright, Gaussian case). Let $\mathbf{g} \sim N(\mathbf{0}, \mathbf{I}_n)$ and let $\mathbf{A}$ be any $n \times n$ matrix. Then for all $t \geq 0$:

$$\mathbb{P}(|\mathbf{g}' \mathbf{A} \mathbf{g} - \text{tr}(\mathbf{A})| \geq t) \leq 2 \exp\left[-c \min\left(\frac{t^2}{\|\mathbf{A}\|_{\text{F}}^2}, \frac{t}{\|\mathbf{A}\|_{\text{op}}}\right)\right], \quad (11)$$

where $c = 1/8$ suffices.

Proof. Since $\mathbf{g}'\mathbf{A}\mathbf{g} = \mathbf{g}'\mathbf{A}_s\mathbf{g}$ and $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}_s)$ for the symmetric part $\mathbf{A}_s = (\mathbf{A} + \mathbf{A}')/2$, while $\|\mathbf{A}_s\|_F \leq \|\mathbf{A}\|_F$ and $\|\mathbf{A}_s\|_{\text{op}} \leq \|\mathbf{A}\|_{\text{op}}$ (as $\|\mathbf{A}'\|_F = \|\mathbf{A}\|_F$, $\|\mathbf{A}'\|_{\text{op}} = \|\mathbf{A}\|_{\text{op}}$) make the right-hand side of (11) only larger, it suffices to prove the bound for $\mathbf{A}_s$; we may thus assume $\mathbf{A}$ is symmetric. Let $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'$ with $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Set $\tilde{\mathbf{g}} = \mathbf{U}'\mathbf{g}$, which is again $N(\mathbf{0}, \mathbf{I}_n)$ since $\mathbf{U}$ is orthogonal. Then

$$\mathbf{g}'\mathbf{A}\mathbf{g} = \tilde{\mathbf{g}}'\mathbf{\Lambda}\tilde{\mathbf{g}} = \sum_{k=1}^n \lambda_k \tilde{g}_k^2,$$

a weighted sum of independent χ_1^2 variables.

MGF. For any μ with $\mu\lambda_k < 1/2$ for all k ,

$$\mathbb{E}[e^{\mu \tilde{g}_k^2}] = \int_{-\infty}^{\infty} \frac{e^{\mu u^2 - u^2/2}}{\sqrt{2\pi}} du = \frac{1}{\sqrt{1 - 2\mu}}.$$

Hence, for $|\mu|\|\mathbf{A}\|_{\text{op}} < 1/2$,

$$\mathbb{E}[e^{\mu(\mathbf{g}'\mathbf{A}\mathbf{g} - \text{tr}(\mathbf{A}))}] = \prod_{k=1}^n \frac{e^{-\mu\lambda_k}}{\sqrt{1 - 2\mu\lambda_k}}. \quad (12)$$

Bound the MGF. For $|u| \leq 1/4$, the elementary inequality $-\ln(1 - 2u) - 2u \leq 4u^2$ holds (Taylor expansion: $-\ln(1 - 2u) - 2u = 2u^2 + \frac{8}{3}u^3 + 4u^4 + \dots \leq 2u^2/(1 - 2|u|) \leq 4u^2$ for $|u| \leq 1/4$). Applying this with $u = \mu\lambda_k$ for all k (which requires $\mu\|\mathbf{A}\|_{\text{op}} \leq 1/4$) and summing the halved version $-\frac{1}{2}\ln(1 - 2u) - u \leq 2u^2$:

$$\sum_{k=1}^n \left[-\frac{1}{2}\ln(1 - 2\mu\lambda_k) - \mu\lambda_k \right] \leq 2\mu^2 \sum_{k=1}^n \lambda_k^2 = 2\mu^2 \|\mathbf{A}\|_F^2,$$

so

$$\mathbb{E}[e^{\mu(\mathbf{g}'\mathbf{A}\mathbf{g} - \text{tr}(\mathbf{A}))}] \leq e^{2\mu^2 \|\mathbf{A}\|_F^2} \quad \text{for } |\mu| \leq \frac{1}{4\|\mathbf{A}\|_{\text{op}}}. \quad (13)$$

Chernoff with constrained optimization. For $t > 0$, Markov's inequality gives

$$\mathbb{P}(\mathbf{g}'\mathbf{A}\mathbf{g} - \text{tr}(\mathbf{A}) \geq t) \leq \inf_{0 < \mu \leq 1/(4\|\mathbf{A}\|_{\text{op}})} \exp(-\mu t + 2\mu^2 \|\mathbf{A}\|_F^2).$$

The unconstrained optimum is $\mu^* = t/(4\|\mathbf{A}\|_F^2)$.

Case 1 (sub-Gaussian regime): $\mu^* \leq 1/(4\|\mathbf{A}\|_{\text{op}})$, i.e., $t \leq \|\mathbf{A}\|_F^2/\|\mathbf{A}\|_{\text{op}}$. Substituting μ^* :

$$\mathbb{P}(\mathbf{g}'\mathbf{A}\mathbf{g} - \text{tr}(\mathbf{A}) \geq t) \leq \exp\left(-\frac{t^2}{8\|\mathbf{A}\|_F^2}\right).$$

Case 2 (sub-exponential regime): $t > \|\mathbf{A}\|_F^2/\|\mathbf{A}\|_{\text{op}}$. Set $\mu = 1/(4\|\mathbf{A}\|_{\text{op}})$:

$$-\mu t + 2\mu^2 \|\mathbf{A}\|_F^2 = -\frac{t}{4\|\mathbf{A}\|_{\text{op}}} + \frac{\|\mathbf{A}\|_F^2}{8\|\mathbf{A}\|_{\text{op}}^2} \leq -\frac{t}{4\|\mathbf{A}\|_{\text{op}}} + \frac{t}{8\|\mathbf{A}\|_{\text{op}}} = -\frac{t}{8\|\mathbf{A}\|_{\text{op}}},$$

using $\|\mathbf{A}\|_F^2/\|\mathbf{A}\|_{\text{op}}^2 \leq t/\|\mathbf{A}\|_{\text{op}}$ in Case 2. Hence

$$\mathbb{P}(\mathbf{g}'\mathbf{A}\mathbf{g} - \text{tr}(\mathbf{A}) \geq t) \leq \exp\left(-\frac{t}{8\|\mathbf{A}\|_{\text{op}}}\right).$$

Combining the two cases gives the one-sided bound with $c = 1/8$. Applying the same argument to $-\mathbf{A}$ and a union bound yields (11). $\square$

Sub-Gaussian case

Theorem 3.3 (Hanson–Wright Inequality). Let $\mathbf{x} = (X_1, \dots, X_n)' \in \mathbb{R}^n$ be a random vector with independent, mean-zero, sub-Gaussian components satisfying $\|X_i\|_{\psi_2} \leq K$. Let $\mathbf{A}$ be an $n \times n$ matrix. Then for all $t \geq 0$:

$$\mathbb{P}(|\mathbf{x}'\mathbf{A}\mathbf{x} - \mathbb{E}[\mathbf{x}'\mathbf{A}\mathbf{x}]| \geq t) \leq 2 \exp \left[-c \min \left(\frac{t^2}{K^4 \|\mathbf{A}\|_{\text{F}}^2}, \frac{t}{K^2 \|\mathbf{A}\|_{\text{op}}} \right) \right], \quad (14)$$

where $c > 0$ is a universal constant.

The proof follows the skeleton of the Gaussian case, with two substantive lemmas inserted: a *decoupling lemma* (Step 2) and an *MGF lemma* for sub-Gaussian quadratic forms (Step 4). Both are stated and proved in full. Nothing else is used beyond the Chernoff method, Bernstein's inequality (Theorem 4.1), the sub-Gaussian MGF characterization (Theorem 1.1(i)), and the Gaussian case just proved (Theorem 3.2). The argument has eight steps.

Proof. Step 1: Reduce to the zero-diagonal case. Write $\mathbf{A} = \mathbf{D} + \mathbf{A}_0$, where $\mathbf{D}$ is the diagonal matrix $\text{diag}(a_{11}, \dots, a_{nn})$ and $\mathbf{A}_0$ has zero diagonal. Then

$$\mathbf{x}'\mathbf{A}\mathbf{x} - \mathbb{E}[\mathbf{x}'\mathbf{A}\mathbf{x}] = \underbrace{\sum_{i=1}^n a_{ii}(X_i^2 - \mathbb{E}[X_i^2])}_{\text{diagonal part}} + \underbrace{\mathbf{x}'\mathbf{A}_0\mathbf{x}}_{\text{off-diagonal part}},$$

since $\mathbb{E}[\mathbf{x}'\mathbf{A}_0\mathbf{x}] = \sum_{i \neq j} (\mathbf{A}_0)_{ij} \mathbb{E}[X_i] \mathbb{E}[X_j] = 0$ by independence and mean zero.

Diagonal part. The variables $Y_i = X_i^2 - \mathbb{E}[X_i^2]$ are independent and mean-zero. By Proposition 4.1, X_i^2 is sub-exponential with $\|X_i^2\|_{\psi_1} \leq C_0 K^2$; subtracting the constant $\mathbb{E}[X_i^2]$ changes the ψ_1 -norm by at most a constant factor, since $|\mathbb{E}[X_i^2]| \leq C_0 K^2$ by Proposition 1.2. Hence $\|Y_i\|_{\psi_1} \leq C_0 K^2$. The summands $a_{ii}Y_i$ then satisfy $\sum_i \|a_{ii}Y_i\|_{\psi_1}^2 \leq C_0^2 K^4 \|\mathbf{D}\|_{\text{F}}^2$ and $\max_i \|a_{ii}Y_i\|_{\psi_1} \leq C_0 K^2 \|\mathbf{D}\|_{\text{op}}$, so Bernstein's inequality (Theorem 4.1) gives

$$\mathbb{P} \left(\left| \sum_{i=1}^n a_{ii}Y_i \right| \geq t \right) \leq 2 \exp \left[-c_1 \min \left(\frac{t^2}{K^4 \|\mathbf{D}\|_{\text{F}}^2}, \frac{t}{K^2 \|\mathbf{D}\|_{\text{op}}} \right) \right], \quad (15)$$

with $\|\mathbf{D}\|_{\text{F}} \leq \|\mathbf{A}\|_{\text{F}}$ and $\|\mathbf{D}\|_{\text{op}} = \max_i |a_{ii}| \leq \|\mathbf{A}\|_{\text{op}}$. It remains to bound the off-diagonal part $\mathbf{x}'\mathbf{A}_0\mathbf{x}$, which occupies Steps 2–7.

Step 2: Decoupling lemma. For a subset $I \subseteq \{1, \dots, n\}$, write I^c for its complement and let $\mathbf{A}_0^{(I)}$ be the *masked matrix* with entries $(\mathbf{A}_0^{(I)})_{ij} = (\mathbf{A}_0)_{ij}$ when $i \in I$ and $j \in I^c$, and $(\mathbf{A}_0^{(I)})_{ij} = 0$ otherwise. Let $\mathbf{y}$ be an independent copy of $\mathbf{x}$.

Decoupling lemma. For every convex function $F : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}[F(\mathbf{x}'\mathbf{A}_0\mathbf{x})] \leq \frac{1}{2^n} \sum_{I \subseteq \{1, \dots, n\}} \mathbb{E} \left[F \left(4 \mathbf{x}'\mathbf{A}_0^{(I)}\mathbf{y} \right) \right]. \quad (16)$$

Moreover, $\|\mathbf{A}_0^{(I)}\|_{\text{F}} \leq \|\mathbf{A}_0\|_{\text{F}}$ and $\|\mathbf{A}_0^{(I)}\|_{\text{op}} \leq \|\mathbf{A}_0\|_{\text{op}}$ for every I .

Proof. Let $\delta_1, \dots, \delta_n$ be independent Bernoulli(1/2) variables, independent of $\mathbf{x}$ and $\mathbf{y}$, and set $I = \{i : \delta_i = 1\}$. For any ordered pair $i \neq j$, independence gives $\mathbb{P}(i \in I, j \in I^c) = \mathbb{P}(\delta_i = 1)\mathbb{P}(\delta_j = 0) = 1/4$. Since $\mathbf{A}_0$ has zero diagonal, $\mathbf{x}'\mathbf{A}_0\mathbf{x} = \sum_{i \neq j} (\mathbf{A}_0)_{ij} X_i X_j$, so, writing $\mathbb{E}_I$ for expectation over the δ_i only,

$$\mathbb{E}_I \left[\sum_{i \in I, j \in I^c} (\mathbf{A}_0)_{ij} X_i X_j \right] = \sum_{i \neq j} (\mathbf{A}_0)_{ij} X_i X_j \cdot \mathbb{P}(i \in I, j \in I^c) = \frac{1}{4} \mathbf{x}'\mathbf{A}_0\mathbf{x}.$$

Thus $\mathbf{x}'\mathbf{A}_0\mathbf{x} = \mathbb{E}_I[4 \sum_{i \in I, j \in I^c} (\mathbf{A}_0)_{ij} X_i X_j]$. Applying Jensen's inequality to the convex F (conditionally on $\mathbf{x}$),

$$F(\mathbf{x}'\mathbf{A}_0\mathbf{x}) \leq \mathbb{E}_I \left[F \left(4 \sum_{i \in I, j \in I^c} (\mathbf{A}_0)_{ij} X_i X_j \right) \right].$$

Take $\mathbb{E}_{\mathbf{x}}$ on both sides; since $\mathbb{E}_I$ is an average over the 2^n subsets, and the convex F is bounded below by an affine function (so each term is well-defined in $(-\infty, +\infty]$), linearity of expectation over this finite sum gives

$$\mathbb{E}_{\mathbf{x}}[F(\mathbf{x}'\mathbf{A}_0\mathbf{x})] \leq \frac{1}{2^n} \sum_I \mathbb{E}_{\mathbf{x}} \left[F \left(4 \sum_{i \in I, j \in I^c} (\mathbf{A}_0)_{ij} X_i X_j \right) \right].$$

Now fix a subset I . The sum $\sum_{i \in I, j \in I^c} (\mathbf{A}_0)_{ij} X_i X_j$ depends on $\mathbf{x}$ only through the two disjoint — hence independent — coordinate blocks $(X_i)_{i \in I}$ and $(X_j)_{j \in I^c}$. Because $\mathbf{y}$ is an independent copy of $\mathbf{x}$, the pair of blocks $((X_i)_{i \in I}, (X_j)_{j \in I^c})$ has the same joint distribution as $((X_i)_{i \in I}, (Y_j)_{j \in I^c})$. Replacing X_j by Y_j for $j \in I^c$ therefore leaves the distribution of the bilinear sum unchanged:

$$\sum_{i \in I, j \in I^c} (\mathbf{A}_0)_{ij} X_i X_j \stackrel{d}{=} \sum_{i \in I, j \in I^c} (\mathbf{A}_0)_{ij} X_i Y_j = \mathbf{x}'\mathbf{A}_0^{(I)}\mathbf{y},$$

and substituting into the previous display yields (16).

For the norm bounds: $\mathbf{A}_0^{(I)}$ is obtained from $\mathbf{A}_0$ by setting some entries to zero, so $\|\mathbf{A}_0^{(I)}\|_{\text{F}}^2 = \sum_{i \in I, j \in I^c} (\mathbf{A}_0)_{ij}^2 \leq \|\mathbf{A}_0\|_{\text{F}}^2$. Writing $\mathbf{P}_I$ for the diagonal 0/1 projection onto the coordinates indexed by I and $\mathbf{P}_{I^c} = \mathbf{I}_n - \mathbf{P}_I$, one has $\mathbf{A}_0^{(I)} = \mathbf{P}_I \mathbf{A}_0 \mathbf{P}_{I^c}$; since $\|\mathbf{P}_I\|_{\text{op}} \leq 1$ and $\|\mathbf{P}_{I^c}\|_{\text{op}} \leq 1$, it follows that $\|\mathbf{A}_0^{(I)}\|_{\text{op}} \leq \|\mathbf{A}_0\|_{\text{op}}$. $\square$

Step 3: Conditional sub-Gaussian MGF of the bilinear form. Fix a subset I and condition on $\mathbf{y}$. Then $\mathbf{x}'\mathbf{A}_0^{(I)}\mathbf{y} = \sum_{i=1}^n X_i (\mathbf{A}_0^{(I)}\mathbf{y})_i$ is, in the randomness of $\mathbf{x}$, a sum of independent mean-zero random variables. Each summand is sub-Gaussian with $\|X_i (\mathbf{A}_0^{(I)}\mathbf{y})_i\|_{\psi_2} = |(\mathbf{A}_0^{(I)}\mathbf{y})_i| \|X_i\|_{\psi_2} \leq K |(\mathbf{A}_0^{(I)}\mathbf{y})_i|$. By the sub-Gaussian MGF characterization (Theorem 1.1(i)), there is a universal constant $C_1 > 0$ with $\mathbb{E}[e^{sX_i (\mathbf{A}_0^{(I)}\mathbf{y})_i} \mid \mathbf{y}] \leq \exp(C_1 K^2 (\mathbf{A}_0^{(I)}\mathbf{y})_i^2 s^2/2)$ for every $s \in \mathbb{R}$. Multiplying over the independent coordinates,

$$\mathbb{E} \left[e^{\lambda \mathbf{x}'\mathbf{A}_0^{(I)}\mathbf{y}} \mid \mathbf{y} \right] = \prod_{i=1}^n \mathbb{E} \left[e^{\lambda X_i (\mathbf{A}_0^{(I)}\mathbf{y})_i} \mid \mathbf{y} \right] \leq \exp \left(\frac{C_1 K^2 \lambda^2 \|\mathbf{A}_0^{(I)}\mathbf{y}\|^2}{2} \right), \quad \lambda \in \mathbb{R}. \quad (17)$$

Step 4: MGF lemma for sub-Gaussian quadratic forms. To take the expectation of (17) over $\mathbf{y}$ we need an MGF bound for the quadratic form $\|\mathbf{A}_0^{(I)}\mathbf{y}\|^2 = \mathbf{y}'(\mathbf{A}_0^{(I)})' \mathbf{A}_0^{(I)}\mathbf{y}$. We prove the following.

MGF lemma. Let $\mathbf{y} \in \mathbb{R}^n$ have independent, mean-zero, sub-Gaussian coordinates with $\|Y_i\|_{\psi_2} \leq K$, and let $\mathbf{B}$ be a symmetric positive semidefinite $n \times n$ matrix. Then there are universal constants $C, c_2 > 0$ such that

$$\mathbb{E} \left[e^{\mu \mathbf{y}'\mathbf{B}\mathbf{y}} \right] \leq \exp(C K^2 \mu \text{tr}(\mathbf{B}) + C^2 K^4 \mu^2 \|\mathbf{B}\|_{\text{F}}^2) \quad \text{for } 0 \leq \mu \leq \frac{c_2}{K^2 \|\mathbf{B}\|_{\text{op}}}. \quad (18)$$

Proof. The argument has three parts.

(1) *Hubbard–Stratonovich identity.* For any $\mathbf{v} \in \mathbb{R}^n$ and $\mathbf{g} \sim N(\mathbf{0}, \mathbf{I}_n)$,

$$\mathbb{E}_{\mathbf{g}}[e^{\langle \mathbf{v}, \mathbf{g} \rangle}] = \prod_{i=1}^n \mathbb{E}[e^{v_i g_i}] = \prod_{i=1}^n e^{v_i^2/2} = e^{\|\mathbf{v}\|^2/2}.$$

Let $\mathbf{B}^{1/2}$ be the symmetric positive semidefinite square root of $\mathbf{B}$. Applying the identity with $\mathbf{v} = \sqrt{2\mu} \mathbf{B}^{1/2}\mathbf{y}$ (here $\mu \geq 0$), and noting $\|\mathbf{v}\|^2/2 = \mu \mathbf{y}'\mathbf{B}^{1/2}\mathbf{B}^{1/2}\mathbf{y} = \mu \mathbf{y}'\mathbf{B}\mathbf{y}$,

$$e^{\mu \mathbf{y}'\mathbf{B}\mathbf{y}} = \mathbb{E}_{\mathbf{g}} \left[e^{\sqrt{2\mu} \mathbf{g}'\mathbf{B}^{1/2}\mathbf{y}} \right]. \quad (19)$$

(2) *Swap the expectations and use the sub-Gaussian MGF of a linear form.* Both sides of (19) are nonnegative, so taking $\mathbb{E}_{\mathbf{y}}$ and applying Tonelli's theorem to exchange the order of integration,

$$\mathbb{E}_{\mathbf{y}} \left[e^{\mu \mathbf{y}' \mathbf{B} \mathbf{y}} \right] = \mathbb{E}_{\mathbf{g}} \mathbb{E}_{\mathbf{y}} \left[e^{\sqrt{2\mu} (\mathbf{B}^{1/2} \mathbf{g})' \mathbf{y}} \right].$$

Fix $\mathbf{g}$ and set $\mathbf{w} = \mathbf{B}^{1/2} \mathbf{g}$. The coordinates Y_i are independent, mean-zero, with $\|Y_i\|_{\psi_2} \leq K$, so by Theorem 1.1(i), $\mathbb{E}[e^{sY_i}] \leq e^{C_1 K^2 s^2/2}$ with the same universal C_1 as in Step 3. Hence

$$\mathbb{E}_{\mathbf{y}} \left[e^{\sqrt{2\mu} \mathbf{w}' \mathbf{y}} \right] = \prod_{i=1}^n \mathbb{E} \left[e^{\sqrt{2\mu} w_i Y_i} \right] \leq \prod_{i=1}^n e^{C_1 K^2 \mu w_i^2} = e^{C_1 K^2 \mu \|\mathbf{w}\|^2} = e^{C_1 K^2 \mu \mathbf{g}' \mathbf{B} \mathbf{g}},$$

using $\|\mathbf{w}\|^2 = \mathbf{g}' \mathbf{B}^{1/2} \mathbf{B}^{1/2} \mathbf{g} = \mathbf{g}' \mathbf{B} \mathbf{g}$. Therefore

$$\mathbb{E}_{\mathbf{y}} \left[e^{\mu \mathbf{y}' \mathbf{B} \mathbf{y}} \right] \leq \mathbb{E}_{\mathbf{g}} \left[e^{C_1 K^2 \mu \mathbf{g}' \mathbf{B} \mathbf{g}} \right]. \quad (20)$$

(3) *Apply the Gaussian quadratic-form MGF.* The right-hand side of (20) is a Gaussian quadratic-form MGF, controlled by inequality (13) of Theorem 3.2. Apply that bound with the matrix $\mathbf{B}$ and parameter $\nu := C_1 K^2 \mu$: provided $\nu \leq 1/(4\|\mathbf{B}\|_{\text{op}})$,

$$\mathbb{E}_{\mathbf{g}} \left[e^{\nu (\mathbf{g}' \mathbf{B} \mathbf{g} - \text{tr}(\mathbf{B}))} \right] \leq e^{2\nu^2 \|\mathbf{B}\|_{\text{F}}^2}, \quad \text{i.e.} \quad \mathbb{E}_{\mathbf{g}} \left[e^{\nu \mathbf{g}' \mathbf{B} \mathbf{g}} \right] \leq \exp(\nu \text{tr}(\mathbf{B}) + 2\nu^2 \|\mathbf{B}\|_{\text{F}}^2).$$

Substituting $\nu = C_1 K^2 \mu$ into (20),

$$\mathbb{E}_{\mathbf{y}} \left[e^{\mu \mathbf{y}' \mathbf{B} \mathbf{y}} \right] \leq \exp(C_1 K^2 \mu \text{tr}(\mathbf{B}) + 2C_1^2 K^4 \mu^2 \|\mathbf{B}\|_{\text{F}}^2),$$

valid when $C_1 K^2 \mu \leq 1/(4\|\mathbf{B}\|_{\text{op}})$, that is, for $0 \leq \mu \leq c_2/(K^2 \|\mathbf{B}\|_{\text{op}})$ with $c_2 := 1/(4C_1)$. Since $\text{tr}(\mathbf{B}) \geq 0$ and $\mu \geq 0$, setting $C := \sqrt{2} C_1$ (so that $C \geq C_1$ and $C^2 = 2C_1^2$) yields (18). $\square$

Step 5: MGF of the decoupled bilinear form. Fix a subset I . Taking $\mathbb{E}_{\mathbf{y}}$ of the conditional bound (17) and writing $\mathbf{B} = (\mathbf{A}_0^{(I)})' \mathbf{A}_0^{(I)}$ — which is symmetric positive semidefinite with $\|\mathbf{A}_0^{(I)} \mathbf{y}\|^2 = \mathbf{y}' \mathbf{B} \mathbf{y}$ —

$$\mathbb{E} \left[e^{\lambda \mathbf{x}' \mathbf{A}_0^{(I)} \mathbf{y}} \right] \leq \mathbb{E}_{\mathbf{y}} \left[e^{\mu \mathbf{y}' \mathbf{B} \mathbf{y}} \right], \quad \mu := \frac{C_1 K^2 \lambda^2}{2} \geq 0.$$

The matrix $\mathbf{B}$ satisfies $\text{tr}(\mathbf{B}) = \|\mathbf{A}_0^{(I)}\|_{\text{F}}^2$ and $\|\mathbf{B}\|_{\text{op}} = \|\mathbf{A}_0^{(I)}\|_{\text{op}}^2$, and, since the eigenvalues of $\mathbf{B}$ are the squared singular values σ_k^2 of $\mathbf{A}_0^{(I)}$,

$$\|\mathbf{B}\|_{\text{F}}^2 = \sum_k \sigma_k^4 \leq \left(\max_k \sigma_k^2 \right) \sum_k \sigma_k^2 = \|\mathbf{A}_0^{(I)}\|_{\text{op}}^2 \|\mathbf{A}_0^{(I)}\|_{\text{F}}^2.$$

The MGF lemma (18) applies when $\mu \leq c_2/(K^2 \|\mathbf{B}\|_{\text{op}})$, i.e. when $C_1 K^2 \lambda^2/2 \leq c_2/(K^2 \|\mathbf{A}_0^{(I)}\|_{\text{op}}^2)$, equivalently $|\lambda| \leq c_3/(K^2 \|\mathbf{A}_0^{(I)}\|_{\text{op}})$ with $c_3 := \sqrt{2c_2/C_1}$. Within this range it gives

$$\mathbb{E} \left[e^{\lambda \mathbf{x}' \mathbf{A}_0^{(I)} \mathbf{y}} \right] \leq \exp \left(\frac{1}{2} C C_1 K^4 \lambda^2 \|\mathbf{A}_0^{(I)}\|_{\text{F}}^2 + \frac{1}{4} C^2 C_1^2 K^8 \lambda^4 \|\mathbf{A}_0^{(I)}\|_{\text{op}}^2 \|\mathbf{A}_0^{(I)}\|_{\text{F}}^2 \right).$$

In the second exponent term, factor out $K^4 \lambda^2 \|\mathbf{A}_0^{(I)}\|_{\text{F}}^2$: the remaining factor $K^4 \lambda^2 \|\mathbf{A}_0^{(I)}\|_{\text{op}}^2$ is at most c_3^2 throughout the admissible range $|\lambda| \leq c_3/(K^2 \|\mathbf{A}_0^{(I)}\|_{\text{op}})$. Hence that term is bounded by $\frac{1}{4} C^2 C_1^2 c_3^2 K^4 \lambda^2 \|\mathbf{A}_0^{(I)}\|_{\text{F}}^2$, and absorbing both terms into the universal constant $C_2 := \frac{1}{2} C C_1 + \frac{1}{4} C^2 C_1^2 c_3^2$,

$$\mathbb{E} \left[e^{\lambda \mathbf{x}' \mathbf{A}_0^{(I)} \mathbf{y}} \right] \leq \exp \left(C_2 K^4 \lambda^2 \|\mathbf{A}_0^{(I)}\|_{\text{F}}^2 \right) \quad \text{for } |\lambda| \leq \frac{c_3}{K^2 \|\mathbf{A}_0^{(I)}\|_{\text{op}}}. \quad (21)$$

Step 6: Decoupling the MGF of $\mathbf{x}'\mathbf{A}_0\mathbf{x}$. Apply the decoupling lemma (16) with the convex function $F(u) = e^{\lambda u}$:

$$\mathbb{E}\left[e^{\lambda \mathbf{x}'\mathbf{A}_0\mathbf{x}}\right] \leq \frac{1}{2^n} \sum_I \mathbb{E}\left[e^{4\lambda \mathbf{x}'\mathbf{A}_0^{(I)}\mathbf{y}}\right].$$

By the norm bounds of Step 2, every I has $\|\mathbf{A}_0^{(I)}\|_{\text{op}} \leq \|\mathbf{A}_0\|_{\text{op}}$ and $\|\mathbf{A}_0^{(I)}\|_{\text{F}} \leq \|\mathbf{A}_0\|_{\text{F}}$. Hence if $|4\lambda| \leq c_3/(K^2\|\mathbf{A}_0\|_{\text{op}})$, then $|4\lambda| \leq c_3/(K^2\|\mathbf{A}_0^{(I)}\|_{\text{op}})$ for every I , and (21) (with λ replaced by 4λ) applies to every term of the average:

$$\mathbb{E}\left[e^{4\lambda \mathbf{x}'\mathbf{A}_0^{(I)}\mathbf{y}}\right] \leq \exp\left(16 C_2 K^4 \lambda^2 \|\mathbf{A}_0^{(I)}\|_{\text{F}}^2\right) \leq \exp\left(16 C_2 K^4 \lambda^2 \|\mathbf{A}_0\|_{\text{F}}^2\right).$$

Every term obeys the same bound, hence so does their average:

$$\mathbb{E}\left[e^{\lambda \mathbf{x}'\mathbf{A}_0\mathbf{x}}\right] \leq \exp\left(C_3 K^4 \lambda^2 \|\mathbf{A}_0\|_{\text{F}}^2\right) \quad \text{for } |\lambda| \leq \frac{c_4}{K^2\|\mathbf{A}_0\|_{\text{op}}}, \quad (22)$$

where $C_3 := 16 C_2$ and $c_4 := c_3/4$.

Step 7: Chernoff bound for the off-diagonal part. The bound (22) has exactly the form of (13) in the Gaussian case, with $2\|\mathbf{A}\|_{\text{F}}^2$ replaced by $C_3 K^4 \|\mathbf{A}_0\|_{\text{F}}^2$ and the radius $1/(4\|\mathbf{A}\|_{\text{op}})$ replaced by $c_4/(K^2\|\mathbf{A}_0\|_{\text{op}})$. The constrained Chernoff optimization is therefore identical to the one in the proof of Theorem 3.2. For $t > 0$,

$$\mathbb{P}(\mathbf{x}'\mathbf{A}_0\mathbf{x} \geq t) \leq \inf_{0 < \lambda \leq c_4/(K^2\|\mathbf{A}_0\|_{\text{op}})} \exp(-\lambda t + C_3 K^4 \lambda^2 \|\mathbf{A}_0\|_{\text{F}}^2),$$

with unconstrained optimum $\lambda^* = t/(2C_3 K^4 \|\mathbf{A}_0\|_{\text{F}}^2)$. If $\lambda^* \leq c_4/(K^2\|\mathbf{A}_0\|_{\text{op}})$, substituting λ^* gives $\exp(-t^2/(4C_3 K^4 \|\mathbf{A}_0\|_{\text{F}}^2))$. Otherwise $t > 2C_3 c_4 K^2 \|\mathbf{A}_0\|_{\text{F}}^2 / \|\mathbf{A}_0\|_{\text{op}}$; setting $\lambda = c_4/(K^2\|\mathbf{A}_0\|_{\text{op}})$ and using $\|\mathbf{A}_0\|_{\text{F}}^2 / \|\mathbf{A}_0\|_{\text{op}}^2 \leq t/(2C_3 c_4 K^2 \|\mathbf{A}_0\|_{\text{op}})$ gives $\exp(-c_4 t/(2K^2\|\mathbf{A}_0\|_{\text{op}}))$. Applying the same argument to $-\mathbf{A}_0$ and a union bound,

$$\mathbb{P}(|\mathbf{x}'\mathbf{A}_0\mathbf{x}| \geq t) \leq 2 \exp\left[-c \min\left(\frac{t^2}{K^4 \|\mathbf{A}_0\|_{\text{F}}^2}, \frac{t}{K^2 \|\mathbf{A}_0\|_{\text{op}}}\right)\right], \quad (23)$$

with $c = \min(1/(4C_3), c_4/2)$.

Step 8: Combine the diagonal and off-diagonal parts. By the triangle inequality and the union bound,

$$\mathbb{P}(|\mathbf{x}'\mathbf{A}\mathbf{x} - \mathbb{E}[\mathbf{x}'\mathbf{A}\mathbf{x}]| \geq t) \leq \mathbb{P}\left(\left|\sum_{i=1}^n a_{ii}Y_i\right| \geq \frac{t}{2}\right) + \mathbb{P}(|\mathbf{x}'\mathbf{A}_0\mathbf{x}| \geq \frac{t}{2}).$$

For the off-diagonal part, $\mathbf{A}_0 = \mathbf{A} - \mathbf{D}$ satisfies $\|\mathbf{A}_0\|_{\text{F}} \leq \|\mathbf{A}\|_{\text{F}}$ (the supports of $\mathbf{A}_0$ and $\mathbf{D}$ are disjoint) and $\|\mathbf{A}_0\|_{\text{op}} \leq \|\mathbf{A}\|_{\text{op}} + \|\mathbf{D}\|_{\text{op}} \leq 2\|\mathbf{A}\|_{\text{op}}$. Substituting the diagonal bound (15) and the off-diagonal bound (23), each with t replaced by $t/2$, and using $\|\mathbf{D}\|_{\text{F}} \leq \|\mathbf{A}\|_{\text{F}}$, $\|\mathbf{D}\|_{\text{op}} \leq \|\mathbf{A}\|_{\text{op}}$, both probabilities are bounded by $2 \exp[-c \min(t^2/(K^4 \|\mathbf{A}\|_{\text{F}}^2), t/(K^2 \|\mathbf{A}\|_{\text{op}}))]$ for a universal $c > 0$ (the factors $1/2$ and the constant from $\|\mathbf{A}_0\|_{\text{op}} \leq 2\|\mathbf{A}\|_{\text{op}}$ are absorbed into c). Their sum has the same form with a possibly smaller universal constant, which is (14). $\square$

Remark (Scope of the proof). The argument is genuinely complete. The decoupling lemma is proved in full (Step 2) via the random-subset identity, Jensen's inequality, and an independent-copy substitution; the MGF lemma is proved in full (Step 4), the Hubbard–Stratonovich identity reducing the sub-Gaussian quadratic-form MGF to the Gaussian one; and the Gaussian quadratic-form MGF is computed explicitly in Theorem 3.2 via the spectral decomposition. The Chernoff machinery and the sub-Gaussian Hoeffding inequality are derived elsewhere in the notes. No external black box is used. The original Hanson–Wright (1971) and Rudelson–Vershynin (2013) proofs take different routes through the moment-comparison principle for sub-Gaussian variables; the Hubbard–Stratonovich shortcut bypasses that machinery entirely.

Remark. When $\mathbf{A}$ is symmetric idempotent (as in the classical theory: $\mathbf{A} = \mathbf{H}$ or $\mathbf{A} = \mathbf{M}$), $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}'\mathbf{A}) = \text{tr}(\mathbf{A}) = \text{rank}(\mathbf{A}) = r$, and $\|\mathbf{A}\|_{\text{op}} = 1$. The Hanson–Wright inequality then gives

$$\mathbb{P}(|\mathbf{x}'\mathbf{A}\mathbf{x} - \mathbb{E}[\mathbf{x}'\mathbf{A}\mathbf{x}]| \geq t) \leq 2 \exp \left[-c \min \left(\frac{t^2}{K^4 r}, \frac{t}{K^2} \right) \right].$$

Compare this with the exact chi-square distribution $\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi_r^2$ under Gaussianity (Theorem 4.3 in the classical notes). The Hanson–Wright bound gives the correct “scale” r (in the Gaussian regime, χ_r^2 has variance $2r$) and extends to sub-Gaussian errors with no distributional assumption.

4 Sub-Exponential Variables and Bernstein’s Inequality

4.1 Definition and Basic Properties

Definition 4.1 (Sub-Exponential Random Variable). A random variable X is called **sub-exponential** if there exist constants $K_1, K_2 > 0$ such that

$$\mathbb{E}[e^{tX}] \leq e^{K_1^2 t^2 / 2} \quad \text{for all } |t| \leq 1/K_2. \quad (24)$$

The **sub-exponential norm** (or ψ_1 -norm) is

$$\|X\|_{\psi_1} = \inf \{ t > 0 : \mathbb{E}[e^{|X|/t}] \leq 2 \}. \quad (25)$$

Proposition 4.1 (Squares of Sub-Gaussians Are Sub-Exponential). If X is sub-Gaussian, then X^2 is sub-exponential with $\|X^2\|_{\psi_1} \leq C\|X\|_{\psi_2}^2$.

Proof. We have $\mathbb{E}[e^{X^2/t}] = \mathbb{E}[e^{|X|^2/t}]$. By the definition of $\|X\|_{\psi_2}$, $\mathbb{E}[e^{X^2/\|X\|_{\psi_2}^2}] \leq 2$. Setting $t = \|X\|_{\psi_2}^2$, we get $\mathbb{E}[e^{|X|^2/t}] \leq 2$. Hence $\|X^2\|_{\psi_1} \leq \|X\|_{\psi_2}^2$. $\square$

Example 4.1.

- (1) χ_1^2 : if $Z \sim N(0, 1)$, then Z^2 is sub-exponential by Proposition 4.1. More generally, $\chi_n^2 = \sum_{i=1}^n Z_i^2$ is sub-exponential.
- (2) $\text{Exp}(\lambda)$: if $X \sim \text{Exp}(\lambda)$ (rate λ), then $\mathbb{E}[e^{X/t}] = \lambda/(\lambda - 1/t) = 2$ at $t = 2/\lambda$, so $\|X\|_{\psi_1} = 2/\lambda$.
- (3) Products: if X, Y are sub-Gaussian, then XY is sub-exponential with $\|XY\|_{\psi_1} \leq \|X\|_{\psi_2} \|Y\|_{\psi_2}$.

4.2 Bernstein’s Inequality

Theorem 4.1 (Bernstein’s Inequality). Let $X_1, \dots, X_n$ be independent, mean-zero, sub-exponential random variables. Then for all $t \geq 0$:

$$\mathbb{P} \left(\left| \sum_{i=1}^n X_i \right| \geq t \right) \leq 2 \exp \left[-c \min \left(\frac{t^2}{\sum_{i=1}^n \|X_i\|_{\psi_1}^2}, \frac{t}{\max_i \|X_i\|_{\psi_1}} \right) \right], \quad (26)$$

where $c > 0$ is a universal constant.

Proof. For $|\lambda| \leq 1/(C \max_i \|X_i\|_{\psi_1})$, each factor satisfies $\mathbb{E}[e^{\lambda X_i}] \leq e^{C'^2 \|X_i\|_{\psi_1}^2 \lambda^2 / 2}$ by the sub-exponential MGF condition. The Chernoff argument gives:

$$\mathbb{P} \left(\sum X_i \geq t \right) \leq \exp \left(-\lambda t + \frac{C'^2 \lambda^2}{2} \sum \|X_i\|_{\psi_1}^2 \right).$$

We optimize subject to the constraint $\lambda \leq 1/(C \max_i \|X_i\|_{\psi_1})$. The unconstrained optimum is $\lambda^* = t/(C'^2 \sum \|X_i\|_{\psi_1}^2)$.

Case 1: λ^* satisfies the constraint, i.e., $t \leq C'^2 \sum \|X_i\|_{\psi_1}^2 / (C \max_i \|X_i\|_{\psi_1})$. Substituting λ^* gives the sub-Gaussian regime: $\exp(-t^2/(2C'^2 \sum \|X_i\|_{\psi_1}^2))$.

Case 2: λ^* violates the constraint. Set $\lambda = 1/(C \max_i \|X_i\|_{\psi_1})$:

$$\mathbb{P}\left(\sum X_i \geq t\right) \leq \exp\left(-\frac{t}{C \max_i \|X_i\|_{\psi_1}} + \frac{C'^2 \sum \|X_i\|_{\psi_1}^2}{2C^2(\max_i \|X_i\|_{\psi_1})^2}\right).$$

Since the constraint is violated, $t > C'^2 \sum \|X_i\|_{\psi_1}^2 / (C \max_i \|X_i\|_{\psi_1})$, so the constant (second) term obeys

$$\frac{C'^2 \sum \|X_i\|_{\psi_1}^2}{2C^2(\max_i \|X_i\|_{\psi_1})^2} = \frac{1}{2C \max_i \|X_i\|_{\psi_1}} \cdot \frac{C'^2 \sum \|X_i\|_{\psi_1}^2}{C \max_i \|X_i\|_{\psi_1}} < \frac{t}{2C \max_i \|X_i\|_{\psi_1}}.$$

Hence the exponent is at most $-\frac{t}{C \max_i \|X_i\|_{\psi_1}} + \frac{t}{2C \max_i \|X_i\|_{\psi_1}} = -\frac{t}{2C \max_i \|X_i\|_{\psi_1}}$, giving $\exp(-ct / \max_i \|X_i\|_{\psi_1})$ with $c = 1/(2C)$.

The minimum in (26) captures both cases. The two-sided bound follows by the union bound. $\square$

Remark (Comparison: Hoeffding vs. Bernstein). For bounded variables $|X_i| \leq M$ (hence sub-Gaussian and sub-exponential), both inequalities apply. Hoeffding gives purely Gaussian tails $\sim e^{-ct^2}$. Bernstein also gives Gaussian tails for small t , but transitions to exponential tails $\sim e^{-ct}$ for $t \gtrsim \sum \|X_i\|_{\psi_1}^2 / \max_i \|X_i\|_{\psi_1}$. Bernstein can be sharper because its Gaussian regime uses the variance $\sum \|X_i\|_{\psi_1}^2$, which can be much smaller than the range-based quantity $\sum (b_i - a_i)^2$ in Hoeffding's bound when the variables are concentrated near zero.

Part II

Random Matrices: Covariance Estimation and Design Conditioning

5 Covering Numbers and ε -Nets

Definition 5.1 (Covering Number and ε -Net). Let (T, d) be a metric space and $\varepsilon > 0$. An ε -**net** of T is a subset $\mathcal{N} \subseteq T$ such that for every $\mathbf{v} \in T$ there exists $\mathbf{u} \in \mathcal{N}$ with $d(\mathbf{v}, \mathbf{u}) \leq \varepsilon$. The **covering number** $N(T, d, \varepsilon)$ is the smallest cardinality of an ε -net of T .

Lemma 5.1 (Covering Number of the Unit Sphere). For every $\varepsilon \in (0, 1)$:

$$N(S^{n-1}, \|\cdot\|, \varepsilon) \leq \left(\frac{2}{\varepsilon} + 1\right)^n. \quad (27)$$

In particular, $N(S^{n-1}, \|\cdot\|, 1/4) \leq 9^n$.

Proof. Let $\mathcal{N}$ be a maximal ε -separated subset of S^{n-1} : $\|\mathbf{u} - \mathbf{v}\| > \varepsilon$ for all distinct $\mathbf{u}, \mathbf{v} \in \mathcal{N}$, and $\mathcal{N}$ cannot be enlarged. Then $\mathcal{N}$ is an ε -net (otherwise some point has distance $> \varepsilon$ from every element of $\mathcal{N}$, contradicting maximality). The open balls $B(\mathbf{u}, \varepsilon/2)$ for $\mathbf{u} \in \mathcal{N}$ are pairwise disjoint (if $\mathbf{w} \in B(\mathbf{u}_1, \varepsilon/2) \cap B(\mathbf{u}_2, \varepsilon/2)$, then $\|\mathbf{u}_1 - \mathbf{u}_2\| \leq \|\mathbf{u}_1 - \mathbf{w}\| + \|\mathbf{w} - \mathbf{u}_2\| < \varepsilon$, contradicting separation). All such balls lie in $B(\mathbf{0}, 1 + \varepsilon/2)$. Comparing volumes ($\text{vol}(B(\mathbf{0}, r)) = c_n r^n$):

$$|\mathcal{N}| \cdot (\varepsilon/2)^n \leq (1 + \varepsilon/2)^n \implies |\mathcal{N}| \leq \left(\frac{2 + \varepsilon}{\varepsilon}\right)^n = \left(\frac{2}{\varepsilon} + 1\right)^n. \quad \square$$

Lemma 5.2 (ε -Net Approximation for Operator Norms). Let $\mathbf{A}$ be a symmetric $n \times n$ matrix and $\mathcal{N}$ a $(1/4)$ -net of S^{n-1} . Then:

$$\|\mathbf{A}\|_{\text{op}} \leq 2 \sup_{\mathbf{v} \in \mathcal{N}} |\mathbf{v}' \mathbf{A} \mathbf{v}|. \quad (28)$$

Proof. Since $\mathbf{A}$ is symmetric, $\|\mathbf{A}\|_{\text{op}} = \sup_{\mathbf{v} \in S^{n-1}} |\mathbf{v}' \mathbf{A} \mathbf{v}|$. Let $\mathbf{v}^*$ achieve this supremum and choose $\mathbf{v}_0 \in \mathcal{N}$ with $\|\mathbf{v}^* - \mathbf{v}_0\| \leq 1/4$. Then:

$$\begin{aligned} |\mathbf{v}^{*'} \mathbf{A} \mathbf{v}^* - \mathbf{v}_0' \mathbf{A} \mathbf{v}_0| &= |\mathbf{v}^{*'} \mathbf{A} (\mathbf{v}^* - \mathbf{v}_0) + (\mathbf{v}^* - \mathbf{v}_0)' \mathbf{A} \mathbf{v}_0| \\ &\leq \|\mathbf{A}\|_{\text{op}} \|\mathbf{v}^* - \mathbf{v}_0\| + \|\mathbf{v}^* - \mathbf{v}_0\| \|\mathbf{A}\|_{\text{op}} \leq 2 \cdot \frac{1}{4} \cdot \|\mathbf{A}\|_{\text{op}} = \frac{1}{2} \|\mathbf{A}\|_{\text{op}}. \end{aligned}$$

Hence $\|\mathbf{A}\|_{\text{op}} = |\mathbf{v}^{*'} \mathbf{A} \mathbf{v}^*| \leq |\mathbf{v}_0' \mathbf{A} \mathbf{v}_0| + \frac{1}{2} \|\mathbf{A}\|_{\text{op}}$, which gives $\|\mathbf{A}\|_{\text{op}} \leq 2 |\mathbf{v}_0' \mathbf{A} \mathbf{v}_0| \leq 2 \sup_{\mathbf{v} \in \mathcal{N}} |\mathbf{v}' \mathbf{A} \mathbf{v}|$. $\square$

6 Covariance Estimation

6.1 Setup

Remark. In the classical linear model, the design matrix $\mathbf{X}$ is treated as fixed and the Gram matrix $\mathbf{X}'\mathbf{X}$ is assumed to have full rank. In the modern setting, the rows $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ of $\mathbf{X}$ are i.i.d. draws from a distribution with mean $\mathbf{0}$ and covariance $\Sigma = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']$. The sample covariance is $\hat{\Sigma} = \frac{1}{n} \mathbf{X}'\mathbf{X} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i'$, and the fundamental question is: *how close is $\hat{\Sigma}$ to Σ ?*

Definition 6.1 (Isotropic Random Vector). A random vector $\mathbf{x} \in \mathbb{R}^p$ is **isotropic** if $\mathbb{E}[\mathbf{x}] = \mathbf{0}$ and $\mathbb{E}[\mathbf{x} \mathbf{x}'] = \mathbf{I}_p$.

Remark. Any mean-zero random vector with invertible covariance Σ can be made isotropic by the transformation $\mathbf{z} = \Sigma^{-1/2} \mathbf{x}$, since $\mathbb{E}[\mathbf{z} \mathbf{z}'] = \Sigma^{-1/2} \Sigma \Sigma^{-1/2} = \mathbf{I}_p$. Working with isotropic vectors simplifies notation; all results extend to general Σ via this whitening transformation.

6.2 Operator Norm Bound for Covariance Estimation

Theorem 6.1 (Covariance Estimation for Sub-Gaussian Designs). Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be independent, isotropic, sub-Gaussian random vectors in $\mathbb{R}^p$ with $\|\mathbf{x}_i\|_{\psi_2} \leq K$. Let $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i'$. There exist universal constants $C, c > 0$ such that for every $\delta \in (0, 1)$, if $n \geq CK^4 p / \delta^2$, then with probability at least $1 - 2 \exp(-c\delta^2 n / K^4)$:

$$\boxed{\|\hat{\Sigma} - \mathbf{I}_p\|_{\text{op}} \leq \delta.} \quad (29)$$

Proof. Step 1: Variational representation. Since $\hat{\Sigma} - \mathbf{I}_p$ is symmetric:

$$\|\hat{\Sigma} - \mathbf{I}_p\|_{\text{op}} = \sup_{\mathbf{v} \in S^{p-1}} |\mathbf{v}'(\hat{\Sigma} - \mathbf{I}_p)\mathbf{v}| = \sup_{\mathbf{v} \in S^{p-1}} \left| \frac{1}{n} \sum_{i=1}^n [(\mathbf{v}'\mathbf{x}_i)^2 - 1] \right|.$$

Step 2: Pointwise concentration. Fix $\mathbf{v} \in S^{p-1}$. The random variable $Z_i = (\mathbf{v}'\mathbf{x}_i)^2 - 1$ has $\mathbb{E}[Z_i] = 0$ (isotropic assumption). Since $\mathbf{v}'\mathbf{x}_i$ is sub-Gaussian with $\|\mathbf{v}'\mathbf{x}_i\|_{\psi_2} \leq K$, by Proposition 4.1, $(\mathbf{v}'\mathbf{x}_i)^2$ is sub-exponential with $\|(\mathbf{v}'\mathbf{x}_i)^2\|_{\psi_1} \leq K^2$. Hence Z_i is sub-exponential with $\|Z_i\|_{\psi_1} \leq CK^2$. By Bernstein's inequality (Theorem 4.1):

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n Z_i\right| \geq t\right) \leq 2 \exp\left[-cn \min\left(\frac{t^2}{K^4}, \frac{t}{K^2}\right)\right]. \quad (30)$$

For $t \leq 1$ (the regime of interest), $\min(t^2/K^4, t/K^2) = t^2/K^4$ when $t \leq K^2$, so the bound is $2 \exp(-cnt^2/K^4)$.

Step 3: Discretization. Let $\mathcal{N}$ be a $(1/4)$ -net of S^{p-1} . By Lemma 5.1, $|\mathcal{N}| \leq 9^p$. By Lemma 5.2, $\|\hat{\Sigma} - \mathbf{I}_p\|_{\text{op}} \leq 2 \sup_{\mathbf{v} \in \mathcal{N}} |\mathbf{v}'(\hat{\Sigma} - \mathbf{I}_p)\mathbf{v}|$.

Step 4: Union bound. Set $t = \delta/2$ in (30). For each $\mathbf{v} \in \mathcal{N}$:

$$\mathbb{P}\left(|\mathbf{v}'(\hat{\Sigma} - \mathbf{I}_p)\mathbf{v}| \geq \delta/2\right) \leq 2 \exp\left(-\frac{cn\delta^2}{4K^4}\right).$$

Taking the union bound over $\mathcal{N}$:

$$\begin{aligned} \mathbb{P}\left(\|\hat{\Sigma} - \mathbf{I}_p\|_{\text{op}} \geq \delta\right) &\leq |\mathcal{N}| \cdot 2 \exp\left(-\frac{cn\delta^2}{4K^4}\right) \leq 2 \cdot 9^p \cdot \exp\left(-\frac{cn\delta^2}{4K^4}\right) \\ &= 2 \exp\left(p \ln 9 - \frac{cn\delta^2}{4K^4}\right). \end{aligned}$$

When $n \geq CK^4 p / \delta^2$ with $C = 8 \ln 9 / c$, the exponent is $\leq -cn\delta^2 / (8K^4)$. Hence $\mathbb{P}(\|\hat{\Sigma} - \mathbf{I}_p\|_{\text{op}} \geq \delta) \leq 2 \exp(-c'\delta^2 n / K^4)$ with $c' = c/8$. $\square$

7 Eigenvalues of $\mathbf{X}'\mathbf{X}/n$: When Is the Design Well-Conditioned?

7.1 From Covariance Estimation to Eigenvalue Bounds

Theorem 7.1 (Eigenvalue Bounds for Sub-Gaussian Designs). Under the conditions of Theorem 6.1, if $n \geq CK^4 p$, then with probability at least $1 - 2 \exp(-cp)$:

$$\boxed{\frac{1}{2} \leq \lambda_{\min}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \leq \lambda_{\max}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \leq \frac{3}{2}.} \quad (31)$$

Equivalently, $\frac{n}{2} \mathbf{I}_p \preceq \mathbf{X}'\mathbf{X} \preceq \frac{3n}{2} \mathbf{I}_p$.

Proof. Apply Theorem 6.1 with $\delta = 1/2$. The sample-size condition becomes $n \geq 4CK^4p$. On the resulting high-probability event, $\|\mathbf{X}'\mathbf{X}/n - \mathbf{I}_p\|_{\text{op}} \leq 1/2$. Since $\|\mathbf{A}\|_{\text{op}} = \max(|\lambda_{\min}(\mathbf{A})|, |\lambda_{\max}(\mathbf{A})|)$ for symmetric $\mathbf{A}$, every eigenvalue λ_j of $\mathbf{X}'\mathbf{X}/n$ satisfies $|\lambda_j - 1| \leq 1/2$, giving $1/2 \leq \lambda_j \leq 3/2$. The probability bound is $1 - 2\exp(-c'n/(4K^4)) \geq 1 - 2\exp(-c'p)$ since $n \geq 4CK^4p$. $\square$

Remark (Connection to the classical full-rank assumption). Theorem 7.1 is the non-asymptotic replacement for the classical assumption (A4), $\text{rank}(\mathbf{X}) = k + 1$. The classical assumption is qualitative (invertibility or not); the modern bound is quantitative: it says *how well-conditioned* $\mathbf{X}'\mathbf{X}$ is. Specifically, $\kappa(\mathbf{X}'\mathbf{X}/n) = \lambda_{\max}/\lambda_{\min} \leq 3$ with high probability whenever $n \gtrsim K^4p$. This is the sample-size condition under which OLS is numerically stable.

7.2 The Condition Number and Its Role

Definition 7.1 (Condition Number). The **condition number** of a positive definite matrix $\mathbf{A}$ is

$$\kappa(\mathbf{A}) = \frac{\lambda_{\max}(\mathbf{A})}{\lambda_{\min}(\mathbf{A})} = \|\mathbf{A}\|_{\text{op}} \cdot \|\mathbf{A}^{-1}\|_{\text{op}}. \quad (32)$$

Proposition 7.1 (Condition Number of $\mathbf{X}'\mathbf{X}$ for General Covariance). Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be i.i.d. sub-Gaussian with covariance Σ (not necessarily $\mathbf{I}_p$). If $n \geq CK^4p$, then with high probability:

$$\kappa\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \leq 3\kappa(\Sigma). \quad (33)$$

Proof. Apply Theorem 7.1 to the whitened vectors $\mathbf{z}_i = \Sigma^{-1/2}\mathbf{x}_i$ (which are isotropic sub-Gaussian). Then $\frac{1}{n} \sum \mathbf{z}_i \mathbf{z}_i' = \Sigma^{-1/2} \hat{\Sigma} \Sigma^{-1/2}$, and the eigenvalues of this matrix lie in $[1/2, 3/2]$. The eigenvalues of $\hat{\Sigma} = \mathbf{X}'\mathbf{X}/n$ are those of $\Sigma^{1/2}(\frac{1}{n} \sum \mathbf{z}_i \mathbf{z}_i')\Sigma^{1/2}$. By the min-max theorem:

$$\lambda_{\max}(\hat{\Sigma}) \leq \frac{3}{2}\lambda_{\max}(\Sigma), \quad \lambda_{\min}(\hat{\Sigma}) \geq \frac{1}{2}\lambda_{\min}(\Sigma).$$

Hence $\kappa(\hat{\Sigma}) \leq (3/2)/(1/2) \cdot \kappa(\Sigma) = 3\kappa(\Sigma)$. $\square$

8 The Proportional Regime $p/n \rightarrow \gamma$

8.1 Motivation: The Marchenko–Pastur Law

Remark. When p is fixed and $n \rightarrow \infty$, the law of large numbers ensures $\hat{\Sigma} \rightarrow \Sigma$ and the design is always well-conditioned for large n . But in modern applications (genomics, finance, natural language), p can be comparable to or even larger than n . The **proportional regime** studies $p, n \rightarrow \infty$ with $p/n \rightarrow \gamma \in (0, \infty)$. The Marchenko–Pastur law describes the asymptotic eigenvalue distribution in this regime.

Theorem 8.1 (Marchenko–Pastur Law (Statement Only)). Let $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ be i.i.d. with $\mathbb{E}[\mathbf{x}_i] = \mathbf{0}$ and $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i'] = \mathbf{I}_p$, with i.i.d. entries having unit variance and finite fourth moment. Let $p/n \rightarrow \gamma \in (0, \infty)$. The empirical spectral distribution of $\hat{\Sigma} = \frac{1}{n} \sum \mathbf{x}_i \mathbf{x}_i'$ converges weakly almost surely to the **Marchenko–Pastur distribution** F_γ with density

$$f_\gamma(x) = \frac{1}{2\pi\gamma x} \sqrt{(\lambda_+ - x)(x - \lambda_-)} \mathbf{1}_{[\lambda_-, \lambda_+]}(x), \quad (34)$$

where $\lambda_{\pm} = (1 \pm \sqrt{\gamma})^2$. When $\gamma > 1$, the distribution additionally has a point mass $(1 - 1/\gamma)$ at zero.

Remark (Implications for $\mathbf{X}'\mathbf{X}$). The Marchenko–Pastur law reveals three qualitatively different regimes:

- (1) **Classical regime** ($\gamma \rightarrow 0$, i.e., $p \ll n$): $\lambda_{\pm} \rightarrow 1$. The eigenvalues of $\hat{\Sigma}$ cluster around 1. The design is well-conditioned; the classical theory applies.

- (2) **Proportional regime** ($\gamma \in (0, 1)$): $\lambda_- = (1 - \sqrt{\gamma})^2 > 0$, $\lambda_+ = (1 + \sqrt{\gamma})^2$. The design is invertible but the condition number $\kappa \approx \lambda_+/\lambda_- = ((1 + \sqrt{\gamma})/(1 - \sqrt{\gamma}))^2 \rightarrow \infty$ as $\gamma \rightarrow 1^-$. This is the “hard” regime where classical methods still work but their performance degrades.
- (3) **Overparametrized regime** ($\gamma > 1$, i.e., $p > n$): $\lambda_- = 0$. The design matrix $\mathbf{X}$ has a nontrivial null space, $\mathbf{X}'\mathbf{X}$ is singular, and the OLS estimator $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ does not exist in the classical sense. Regularization (ridge, lasso) or the Moore–Penrose pseudoinverse is needed.

8.2 Non-Asymptotic Bounds in the Proportional Regime

Theorem 8.2 (Extreme Eigenvalue Bounds, Proportional Regime). Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be independent, isotropic, sub-Gaussian random vectors in $\mathbb{R}^p$ with $\|\mathbf{x}_i\|_{\psi_2} \leq K$. There exist constants $C, c > 0$ depending only on K such that for every $t \geq 0$, with probability at least $1 - 2\exp(-ct^2)$:

$$\sqrt{n} - C\sqrt{p} - t \leq \sigma_{\min}(\mathbf{X}) \leq \sigma_{\max}(\mathbf{X}) \leq \sqrt{n} + C\sqrt{p} + t. \quad (35)$$

In terms of $\mathbf{X}'\mathbf{X}/n$:

$$\left(1 - C\sqrt{\frac{p}{n}} - \frac{t}{\sqrt{n}}\right)^2 \leq \lambda_{\min}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \leq \lambda_{\max}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \leq \left(1 + C\sqrt{\frac{p}{n}} + \frac{t}{\sqrt{n}}\right)^2. \quad (36)$$

Proof. We prove the sharp bound for a standard Gaussian matrix—the prototype that fixes the constants—and then state the sub-Gaussian extension.

Gaussian case. Suppose first that the entries of $\mathbf{X}$ are i.i.d. $N(0, 1)$. Write the extreme singular values variationally:

$$\sigma_{\max}(\mathbf{X}) = \max_{\mathbf{u} \in S^{p-1}} \max_{\mathbf{v} \in S^{n-1}} \mathbf{v}'\mathbf{X}\mathbf{u}, \quad \sigma_{\min}(\mathbf{X}) = \min_{\mathbf{u} \in S^{p-1}} \max_{\mathbf{v} \in S^{n-1}} \mathbf{v}'\mathbf{X}\mathbf{u}.$$

Consider the centred Gaussian process $X_{\mathbf{u},\mathbf{v}} = \mathbf{v}'\mathbf{X}\mathbf{u}$ and the comparison process $Y_{\mathbf{u},\mathbf{v}} = \langle \mathbf{g}, \mathbf{v} \rangle + \langle \mathbf{h}, \mathbf{u} \rangle$, with $\mathbf{g} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{h} \sim N(\mathbf{0}, \mathbf{I}_p)$ independent. Since $\mathbb{E}[X_{\mathbf{u},\mathbf{v}}X_{\mathbf{u}',\mathbf{v}'}] = \langle \mathbf{u}, \mathbf{u}' \rangle \langle \mathbf{v}, \mathbf{v}' \rangle$ and $\mathbb{E}[Y_{\mathbf{u},\mathbf{v}}Y_{\mathbf{u}',\mathbf{v}'}] = \langle \mathbf{u}, \mathbf{u}' \rangle + \langle \mathbf{v}, \mathbf{v}' \rangle$, the increments satisfy

$$\mathbb{E}(X_{\mathbf{u},\mathbf{v}} - X_{\mathbf{u}',\mathbf{v}'})^2 = 2 - 2\langle \mathbf{u}, \mathbf{u}' \rangle \langle \mathbf{v}, \mathbf{v}' \rangle, \quad \mathbb{E}(Y_{\mathbf{u},\mathbf{v}} - Y_{\mathbf{u}',\mathbf{v}'})^2 = \|\mathbf{u} - \mathbf{u}'\|^2 + \|\mathbf{v} - \mathbf{v}'\|^2,$$

so that

$$\mathbb{E}(Y_{\mathbf{u},\mathbf{v}} - Y_{\mathbf{u}',\mathbf{v}'})^2 - \mathbb{E}(X_{\mathbf{u},\mathbf{v}} - X_{\mathbf{u}',\mathbf{v}'})^2 = 2(1 - \langle \mathbf{u}, \mathbf{u}' \rangle)(1 - \langle \mathbf{v}, \mathbf{v}' \rangle) \geq 0, \quad (37)$$

with equality whenever $\mathbf{u} = \mathbf{u}'$.

Upper bound. The inequality (37) for *all* pairs is the hypothesis of the Sudakov–Fernique inequality, giving $\mathbb{E} \max_{\mathbf{u},\mathbf{v}} X_{\mathbf{u},\mathbf{v}} \leq \mathbb{E} \max_{\mathbf{u},\mathbf{v}} Y_{\mathbf{u},\mathbf{v}}$. Hence

$$\mathbb{E} \sigma_{\max}(\mathbf{X}) \leq \mathbb{E} \left[\max_{\mathbf{v}} \langle \mathbf{g}, \mathbf{v} \rangle \right] + \mathbb{E} \left[\max_{\mathbf{u}} \langle \mathbf{h}, \mathbf{u} \rangle \right] = \mathbb{E} \|\mathbf{g}\| + \mathbb{E} \|\mathbf{h}\| \leq \sqrt{n} + \sqrt{p},$$

using $\mathbb{E} \|\mathbf{g}\| \leq (\mathbb{E} \|\mathbf{g}\|^2)^{1/2} = \sqrt{n}$ and $\mathbb{E} \|\mathbf{h}\| \leq \sqrt{p}$.

Lower bound. The equality of increments for $\mathbf{u} = \mathbf{u}'$ together with (37) is exactly the hypothesis of **Gordon’s inequality** (the Gaussian min–max comparison theorem; Gordon, 1985): under these conditions $\mathbb{E} \min_{\mathbf{u}} \max_{\mathbf{v}} X_{\mathbf{u},\mathbf{v}} \geq \mathbb{E} \min_{\mathbf{u}} \max_{\mathbf{v}} Y_{\mathbf{u},\mathbf{v}}$. Here

$$\min_{\mathbf{u}} \max_{\mathbf{v}} Y_{\mathbf{u},\mathbf{v}} = \max_{\mathbf{v}} \langle \mathbf{g}, \mathbf{v} \rangle + \min_{\mathbf{u}} \langle \mathbf{h}, \mathbf{u} \rangle = \|\mathbf{g}\| - \|\mathbf{h}\|,$$

so $\mathbb{E} \sigma_{\min}(\mathbf{X}) \geq \mathbb{E} \|\mathbf{g}\| - \mathbb{E} \|\mathbf{h}\| \geq \sqrt{n-1} - \sqrt{p}$, where $\mathbb{E} \|\mathbf{g}\| \geq \sqrt{n-1}$ follows from $(\mathbb{E} \|\mathbf{g}\|)^2 = \mathbb{E} \|\mathbf{g}\|^2 - \text{var} \|\mathbf{g}\| \geq n-1$ (the Gaussian Poincaré inequality gives $\text{var} \|\mathbf{g}\| \leq 1$, as $\mathbf{g} \mapsto \|\mathbf{g}\|$ is 1-Lipschitz).

Concentration. The maps $\mathbf{X} \mapsto \sigma_{\max}(\mathbf{X})$ and $\mathbf{X} \mapsto \sigma_{\min}(\mathbf{X})$ are 1-Lipschitz in the Frobenius norm (singular values are 1-Lipschitz by Weyl’s inequality, and $\|\cdot\|_F \geq \|\cdot\|_{\text{op}}$). Gaussian concentration of

Lipschitz functions gives $\mathbf{P}(|f(\mathbf{X}) - \mathbf{E}f(\mathbf{X})| \geq t) \leq 2e^{-t^2/2}$ for each. With the expectation bounds and a union bound, with probability at least $1 - 2e^{-t^2/2}$,

$$\sqrt{n-1} - \sqrt{p} - t \leq \sigma_{\min}(\mathbf{X}) \leq \sigma_{\max}(\mathbf{X}) \leq \sqrt{n} + \sqrt{p} + t,$$

which is (35) with $C = 1$ (the lower-order gap $\sqrt{n} - \sqrt{n-1} \leq 1$ is absorbed into the additive term).

Sub-Gaussian case. For general isotropic sub-Gaussian rows with $\|\mathbf{x}_i\|_{\psi_2} \leq K$, the same two-sided bound holds with C depending only on K . This is the *matrix deviation inequality* (Vershynin, *HDP*, Theorem 9.1.1; and Bai–Yin, 1993, for the hard edge under finite fourth moments), whose proof replaces the Gaussian comparison by a generic-chaining bound on the sub-Gaussian process $\mathbf{u} \mapsto \|\mathbf{X}\mathbf{u}\| - \sqrt{n}$ over S^{p-1} ; this chaining argument is beyond our scope, and the Gaussian computation above is the prototype that fixes the sharp edges $(1 \pm \sqrt{p/n})^2$ in (36). $\square$

Remark. Setting $t = 0$ in (35), the singular values of $\mathbf{X}$ concentrate around $[\sqrt{n} - C\sqrt{p}, \sqrt{n} + C\sqrt{p}]$, and those of $\mathbf{X}'\mathbf{X}/n$ around $[(1 - C\sqrt{p/n})^2, (1 + C\sqrt{p/n})^2]$. This is the non-asymptotic counterpart of the Marchenko–Pastur support $[\lambda_-, \lambda_+] = [(1 - \sqrt{\gamma})^2, (1 + \sqrt{\gamma})^2]$.

Part III

The Linear Model Redone

9 The Modern Setup

Definition 9.1 (The Linear Model, Modern Version). The data follow

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{X} \in \mathbb{R}^{n \times p}, \quad (38)$$

where p replaces $k + 1$ for the total number of regressors. The following assumptions are imposed:

- (M1) **Random design:** The rows $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ of $\mathbf{X}$ are i.i.d. random vectors with $\mathbb{E}[\mathbf{x}_i] = \mathbf{0}$ and $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i'] = \boldsymbol{\Sigma}$, where $\boldsymbol{\Sigma} \succ 0$.
- (M2) **Sub-Gaussian design:** Each $\mathbf{x}_i$ is sub-Gaussian with $\|\mathbf{x}_i\|_{\psi_2} \leq K$.
- (M3) **Sub-Gaussian noise:** The errors $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d., mean-zero, sub-Gaussian with $\|\varepsilon_i\|_{\psi_2} \leq \sigma$, and independent of $\mathbf{X}$.
- (M4) **Sample size:** $n \geq C_0 K^4 p$ for a sufficiently large universal constant C_0 .

Remark (Comparison with the classical assumptions).

- (1) (M1) replaces the fixed-design assumption. The design is now stochastic.
- (2) (M2) replaces (A4): it is the sub-Gaussian analogue of the full-rank condition. By Theorem 7.1, (M2) and (M4) together imply $\mathbf{X}'\mathbf{X}$ is invertible with high probability.
- (3) (M3) replaces (A5): sub-Gaussian errors instead of normal errors. We write $\sigma_\varepsilon^2 = \text{var}(\varepsilon_i)$ for the error variance; by Proposition 1.2, $\sigma_\varepsilon^2 \leq C \|\varepsilon_i\|_{\psi_2}^2 \leq C \sigma^2$, and throughout σ is treated as a tight sub-Gaussian proxy for the noise scale, so that $\sigma_\varepsilon^2 \asymp \sigma^2$.
- (4) (M4) is the modern analogue of $n > k + 1$ in (A4). It is a quantitative sample-size condition.

10 OLS Prediction Error

Theorem 10.1 (Non-Asymptotic Bound on Prediction Error). Under assumptions (M1)–(M4), with probability at least $1 - 2 \exp(-cp)$:

$$\boxed{\frac{1}{n} \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \leq C \sigma^2 \frac{p}{n}}. \quad (39)$$

Proof. Step 1: Express the prediction error. The OLS estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ (exactly as in the classical derivation, Equation (2.6) in the classical notes). The prediction error is:

$$\frac{1}{n} \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \frac{1}{n} \|\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\|^2 = \frac{1}{n} \|\mathbf{H}\boldsymbol{\varepsilon}\|^2 = \frac{1}{n} \boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}, \quad (40)$$

where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the hat matrix.

Step 2: Conditional expectation. Conditional on $\mathbf{X}$ (and hence on $\mathbf{H}$), the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent sub-Gaussian with mean zero and $\|\varepsilon_i\|_{\psi_2} \leq \sigma$. By the Hanson–Wright inequality (Theorem 3.3) applied conditionally on $\mathbf{X}$:

$$\mathbb{P}(|\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} - \mathbb{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}]| \geq t \mid \mathbf{X}) \leq 2 \exp \left[-c \min \left(\frac{t^2}{\sigma^4 \|\mathbf{H}\|_{\text{F}}^2}, \frac{t}{\sigma^2 \|\mathbf{H}\|_{\text{op}}} \right) \right].$$

Since $\mathbf{H}$ is idempotent with $\text{rank}(\mathbf{H}) = p$ (under the event that $\mathbf{X}$ has full rank):

- $\|\mathbf{H}\|_{\text{op}} = 1$ (eigenvalues are 0 and 1).
- $\|\mathbf{H}\|_{\text{F}}^2 = \text{tr}(\mathbf{H}) = p$.
- $\mathbb{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma_{\varepsilon}^2 \text{tr}(\mathbf{H}) = \sigma_{\varepsilon}^2 p$, where $\sigma_{\varepsilon}^2 = \text{var}(\varepsilon_i) \leq C\sigma^2$.

Here $\sigma_{\varepsilon}^2 = \mathbb{E}[\varepsilon_i^2]$ is the actual error variance. The Hanson–Wright bound gives, for $t = \delta\sigma^2 p$ with $\delta \in (0, 1)$:

$$\mathbb{P}(|\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} - \sigma_{\varepsilon}^2 p| \geq \delta\sigma^2 p \mid \mathbf{X}) \leq 2 \exp[-c \min(\delta^2 p, \delta p)] \leq 2 \exp(-c'\delta^2 p).$$

Step 3: Combine. Setting $\delta = 1$ (say), the event $\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \leq 2\sigma^2 p$ holds with probability $\geq 1 - 2 \exp(-cp)$ conditional on $\mathbf{X}$, provided $\mathbf{X}$ has full rank. By Theorem 7.1, $\mathbf{X}$ has full rank with probability $\geq 1 - 2 \exp(-cp)$ under (M4). By the union bound over both events:

$$\frac{1}{n} \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \frac{1}{n} \boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \leq \frac{2\sigma^2 p}{n} = C\sigma^2 \frac{p}{n}$$

with probability $\geq 1 - 4 \exp(-cp)$. $\square$

Remark. The bound $\frac{1}{n} \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \lesssim \sigma^2 p/n$ says that the **in-sample prediction error** scales linearly with p/n . This is a non-asymptotic version of the well-known heuristic: “each estimated parameter costs you σ^2/n in prediction accuracy.” In the classical theory, the exact result (under normality) is $\mathbb{E}[\frac{1}{n} \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2] = \sigma^2 p/n$. The modern bound matches this rate but holds with high probability and under sub-Gaussian (not Gaussian) errors.

11 OLS Estimation Error

Theorem 11.1 (Non-Asymptotic Bound on Estimation Error). Under assumptions (M1)–(M4), with probability at least $1 - C \exp(-cp)$:

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \leq \frac{C\sigma^2 p}{n \lambda_{\min}(\boldsymbol{\Sigma})}. \quad (41)$$

More precisely, on the high-probability event from Theorem 7.1:

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})' \mathbf{I}_p (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \leq \frac{2}{n \lambda_{\min}(\boldsymbol{\Sigma})} \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \leq \frac{C'\sigma^2 p}{n \lambda_{\min}(\boldsymbol{\Sigma})}. \quad (42)$$

Proof. Step 1: Relate estimation error to prediction error. For any $\mathbf{v} \in \mathbb{R}^p$:

$$\frac{1}{n} \|\mathbf{X}\mathbf{v}\|^2 = \mathbf{v}' \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right) \mathbf{v} \geq \lambda_{\min} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right) \|\mathbf{v}\|^2.$$

Hence $\|\mathbf{v}\|^2 \leq \frac{1}{\lambda_{\min}(\mathbf{X}'\mathbf{X}/n)} \cdot \frac{1}{n} \|\mathbf{X}\mathbf{v}\|^2$. Applying this to $\mathbf{v} = \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$:

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \leq \frac{1}{\lambda_{\min}(\mathbf{X}'\mathbf{X}/n)} \cdot \frac{1}{n} \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2. \quad (43)$$

Step 2: Bound $\lambda_{\min}(\mathbf{X}'\mathbf{X}/n)$ from below. By Theorem 7.1 applied to the whitened design $\mathbf{Z} = \mathbf{X}\boldsymbol{\Sigma}^{-1/2}$, on the high-probability event:

$$\lambda_{\min} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right) \geq \frac{1}{2} \lambda_{\min}(\boldsymbol{\Sigma}).$$

Step 3: Combine. Substituting the prediction error bound (Theorem 10.1) and the eigenvalue bound into (43):

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \leq \frac{2}{\lambda_{\min}(\boldsymbol{\Sigma})} \cdot C\sigma^2 \frac{p}{n} = \frac{C'\sigma^2 p}{n \lambda_{\min}(\boldsymbol{\Sigma})}.$$

The probability is at least $1 - C'' \exp(-cp)$ by the union bound over the two high-probability events. $\square$

Remark (Dependence on the condition number). The estimation error bound (41) involves $1/\lambda_{\min}(\Sigma)$, which equals $\|\Sigma^{-1}\|_{\text{op}}$. When the population covariance is well-conditioned ($\kappa(\Sigma)$ small), $\lambda_{\min}(\Sigma) \approx \lambda_{\max}(\Sigma) \approx \|\Sigma\|_{\text{op}}$, and the bound is comparable to $\sigma^2 p/(n\|\Sigma\|_{\text{op}})$. But when Σ is ill-conditioned (e.g., near-multicollinearity), $\lambda_{\min}(\Sigma) \rightarrow 0$ and the estimation error explodes. This is the modern analogue of the classical observation that near-multicollinearity inflates $\text{var}(\hat{\beta}_j) = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$.

12 Concentration of the Residual Sum of Squares

In the classical theory, $\text{SSE}/\sigma^2 \sim \chi_{n-p}^2$ under normality (Theorem 4.5 in the classical notes). The following theorem is its non-asymptotic, sub-Gaussian counterpart.

Theorem 12.1 (Concentration of SSE). Under assumptions (M1)–(M4) there are universal constants $C', c, c_0 > 0$ such that for every $\eta \in [1, \sqrt{n-p}]$, with probability at least $1 - 2e^{-c\eta^2} - 2e^{-c_0 p}$,

$$\left| \frac{\text{SSE}}{\sigma_\varepsilon^2} - (n-p) \right| \leq C' \eta \sqrt{n-p}, \quad (44)$$

where $\sigma_\varepsilon^2 = \text{var}(\varepsilon_i)$. Consequently $\text{MSE} = \text{SSE}/(n-p)$ satisfies $|\text{MSE} - \sigma_\varepsilon^2| \leq C' \eta \sigma_\varepsilon^2 / \sqrt{n-p}$ on the same event, so at any fixed confidence the deviation is of the central-limit scale $\sqrt{n-p}$ and $\text{MSE} \rightarrow \sigma_\varepsilon^2$ at the $1/\sqrt{n-p}$ rate. (This is the correct scaling: a deviation of order $\sqrt{n-p}$ holds only at constant confidence, exactly as $\text{SSE}/\sigma^2 \sim \chi_{n-p}^2$ has standard deviation $\sqrt{2(n-p)}$ under normality; an exponential-in- p confidence $1 - e^{-cp}$ would require the larger deviation of order $\sqrt{p(n-p)}$, a constant relative error in the proportional regime $p \asymp n$.)

Proof. Step 1: Express SSE as a quadratic form. We have $\text{SSE} = \mathbf{e}'\mathbf{e} = (\mathbf{M}\boldsymbol{\varepsilon})'(\mathbf{M}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$, where $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$ is the residual maker. On the event that $\mathbf{X}$ has full column rank (which holds with probability $\geq 1 - 2\exp(-c_0 p)$ by Theorem 7.1), $\mathbf{M}$ is symmetric idempotent with $\text{tr}(\mathbf{M}) = n-p$, $\|\mathbf{M}\|_{\text{op}} = 1$, and $\|\mathbf{M}\|_{\text{F}}^2 = \text{tr}(\mathbf{M}) = n-p$.

Step 2: Conditional expectation. By the quadratic form expectation lemma (Lemma 3.7 of the classical notes), with $\mathbb{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$ and $\text{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma_\varepsilon^2 \mathbf{I}_n$:

$$\mathbb{E}[\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma_\varepsilon^2 \text{tr}(\mathbf{M}) + \mathbf{0}'\mathbf{M}\mathbf{0} = \sigma_\varepsilon^2(n-p).$$

Step 3: Concentration via Hanson–Wright. Conditional on $\mathbf{X}$, the vector $\boldsymbol{\varepsilon}$ has independent, mean-zero, sub-Gaussian components with $\|\varepsilon_i\|_{\psi_2} \leq \sigma$. By the Hanson–Wright inequality (Theorem 3.3):

$$\mathbb{P}(|\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon} - \sigma_\varepsilon^2(n-p)| \geq t \mid \mathbf{X}) \leq 2 \exp \left[-c \min \left(\frac{t^2}{\sigma^4(n-p)}, \frac{t}{\sigma^2} \right) \right].$$

Here we used $\|\mathbf{M}\|_{\text{F}}^2 = n-p$ and $\|\mathbf{M}\|_{\text{op}} = 1$.

Step 4: Choose t . Apply the bound at the sub-Gaussian scale, $t = \delta \sigma^2(n-p)$ with $\delta \in (0, 1]$. Then the first argument of the minimum is $t^2/(\sigma^4(n-p)) = \delta^2(n-p)$ and the second is $t/\sigma^2 = \delta(n-p) \geq \delta^2(n-p)$, so the minimum equals $\delta^2(n-p)$ and

$$\mathbb{P}(|\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon} - \sigma_\varepsilon^2(n-p)| \geq \delta \sigma^2(n-p) \mid \mathbf{X}) \leq 2 \exp(-c\delta^2(n-p)).$$

Choosing $\delta = C_0/\sqrt{n-p}$ gives $|\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon} - \sigma_\varepsilon^2(n-p)| \leq C_0 \sigma^2 \sqrt{n-p}$ with conditional probability $\geq 1 - 2\exp(-cC_0^2)$. Since the sub-Gaussian proxy is comparable to the noise standard deviation — $\sigma_\varepsilon^2 \asymp \sigma^2$ (Proposition 1.2 gives $\sigma_\varepsilon^2 \leq C\sigma^2$; we take σ to be a tight proxy, so also $\sigma^2 \leq C\sigma_\varepsilon^2$) — dividing by σ_ε^2 yields

$$\left| \frac{\text{SSE}}{\sigma_\varepsilon^2} - (n-p) \right| \leq C'_0 \sqrt{n-p}.$$

Step 5: Remove conditioning. By the union bound over the full-rank event and the concentration event:

$$\mathbb{P} \left(\left| \frac{\text{SSE}}{\sigma_\varepsilon^2} - (n-p) \right| \leq C_0 \sqrt{n-p} \right) \geq 1 - 2\exp(-c''C_0^2) - 2\exp(-c_0 p).$$

This is (44) with $\eta := C_0$ (any value in $[1, \sqrt{n-p}]$ keeps $\delta = C_0/\sqrt{n-p} \leq 1$, the range used in Step 4). The confidence is governed by the deviation parameter η , not by p : at the central-limit scale $\eta = O(1)$ the failure probability $2e^{-c\eta^2}$ is a small tunable constant, and forcing it down to e^{-cp} would require $\eta \asymp \sqrt{p}$, i.e. the larger deviation $\asymp \sqrt{p(n-p)}$. Dividing by $n-p$: $|\text{MSE} - \sigma_\varepsilon^2| \leq C'\eta\sigma_\varepsilon^2/\sqrt{n-p}$. $\square$

13 Non-Asymptotic Confidence Intervals Without Normality

Theorem 13.1 (Sub-Gaussian Confidence Interval for β_j). Under assumptions (M1)–(M4), for each $j \in \{1, \dots, p\}$ and $\alpha \in (0, 1)$: conditionally on $\mathbf{X}$ having full rank, with probability at least $1 - \alpha$,

$$|\hat{\beta}_j - \beta_j| \leq C\sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} \cdot \ln \frac{2}{\alpha}}. \quad (45)$$

Proof. Step 1: Isolate $\hat{\beta}_j - \beta_j$. Let $\mathbf{e}_j \in \mathbb{R}^p$ be the j -th standard basis vector. Then:

$$\hat{\beta}_j - \beta_j = \mathbf{e}_j'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} = \mathbf{c}'\boldsymbol{\varepsilon},$$

where $\mathbf{c} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j \in \mathbb{R}^n$.

Step 2: Conditional sub-Gaussian bound. Fix $\mathbf{X}$ (hence $\mathbf{c}$ is fixed). Then $\mathbf{c}'\boldsymbol{\varepsilon} = \sum_{i=1}^n c_i \varepsilon_i$ is a sum of independent, mean-zero, sub-Gaussian random variables with $\|c_i \varepsilon_i\|_{\psi_2} \leq |c_i|\sigma$. By Theorem 3.1:

$$\mathbb{P}(|\mathbf{c}'\boldsymbol{\varepsilon}| \geq t \mid \mathbf{X}) \leq 2 \exp\left(-\frac{ct^2}{\sigma^2 \|\mathbf{c}\|^2}\right).$$

Step 3: Compute $\|\mathbf{c}\|^2$.

$$\|\mathbf{c}\|^2 = \mathbf{c}'\mathbf{c} = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j = [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}.$$

Step 4: Invert the tail bound. Setting $2 \exp(-ct^2/(\sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj})) = \alpha$ and solving:

$$t = \sigma \sqrt{\frac{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}{c} \cdot \ln \frac{2}{\alpha}} = C'\sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} \cdot \ln \frac{2}{\alpha}},$$

where $C' = 1/\sqrt{c}$. Hence $|\hat{\beta}_j - \beta_j| \leq t$ with probability $\geq 1 - \alpha$, conditional on $\mathbf{X}$. $\square$

Remark (Comparison with the classical interval). The classical t -based confidence interval (Theorem 6.1 of the classical notes) has half-width $t_{\alpha/2, n-p} \cdot s \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$, where $s = \sqrt{\text{MSE}}$. For large $n-p$, the quantile $t_{\alpha/2, n-p} \approx z_{\alpha/2}$, and the Gaussian tail approximation gives $z_{\alpha/2} \approx \sqrt{2 \ln(2/\alpha)}$. So the modern and classical widths agree up to a constant factor. The modern interval has two advantages: (1) no normality assumption on $\boldsymbol{\varepsilon}$, and (2) finite-sample validity. Its disadvantage is that it uses the unknown σ rather than the data-driven s . This can be remedied: by Theorem 12.1, $s^2 = \text{MSE}$ satisfies $|s^2/\sigma_\varepsilon^2 - 1| \leq C/\sqrt{n-p}$ with high probability, so for large $n-p$ one may replace σ by $s\sqrt{2}$ (say) at the cost of doubling the constant and adding the failure probability of the MSE concentration event.

14 Comparison: Classical vs. Modern

Remark (Side-by-side comparison). The following table compares the classical and modern treatments of the linear model. The notation $p = k + 1$.

Aspect	Classical Theory	Modern Theory
Design	Fixed ($\mathbf{X}$ non-random)	Random, i.i.d. rows
Error distribution	$\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$	Sub-Gaussian, $\ \varepsilon_i\ _{\psi_2} \leq \sigma$
Invertibility	$\text{rank}(\mathbf{X}) = p$ (assumed)	$n \geq CK^4 p$ (implies w.h.p.)
Prediction error	$\mathbb{E}[\ \mathbf{H}\boldsymbol{\varepsilon}\ ^2/n] = \sigma^2 p/n$ (exact)	$\ \mathbf{H}\boldsymbol{\varepsilon}\ ^2/n \leq C\sigma^2 p/n$ (w.h.p.)
Estimation error	$\text{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$	$\ \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\ ^2 \leq C\sigma^2 p/(n\lambda_{\min}(\boldsymbol{\Sigma}))$ (w.h.p.)
Distribution of SSE	$\text{SSE}/\sigma^2 \sim \chi_{n-p}^2$	$ \text{SSE}/\sigma^2 - (n-p) \leq C\sqrt{n-p}$ (w.h.p.)
Inference	t, F, χ^2 pivots	Sub-Gaussian concentration (Thm. 13.1)
Regime	p fixed, $n \rightarrow \infty$	$p/n \rightarrow \gamma \in (0, 1)$ allowed

15 What Happens When p Is Comparable to n

Remark. As $p/n \rightarrow \gamma \in (0, 1)$, the OLS estimator remains well-defined but its behavior changes qualitatively.

Theorem 15.1 (Prediction and Estimation Error in the Proportional Regime). Under assumptions (M1)–(M3) with $\boldsymbol{\Sigma} = \mathbf{I}_p$ (isotropic design), conditional on $\mathbf{X}$ having full column rank:

- (i) **In-sample prediction error:** $\mathbb{E}[\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 p/n$, where $\sigma_\varepsilon^2 = \text{var}(\varepsilon_i)$.
- (ii) **Unbiasedness of MSE:** $\mathbb{E}[\text{MSE} \mid \mathbf{X}] = \sigma_\varepsilon^2$.
- (iii) **Out-of-sample prediction risk:** For a new $\mathbf{x}_0$ drawn independently from the same distribution,

$$\mathbb{E}[(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 \text{tr}((\mathbf{X}'\mathbf{X})^{-1}). \quad (46)$$

As $p/n \rightarrow \gamma \in (0, 1)$ with i.i.d. entries having finite fourth moment, $\frac{1}{p} \text{tr}((\mathbf{X}'\mathbf{X}/n)^{-1}) \rightarrow \frac{1}{1-\gamma}$ almost surely by the Marchenko–Pastur law (Theorem 8.1), giving $\mathbb{E}[(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 \mid \mathbf{X}] \rightarrow \sigma_\varepsilon^2 \cdot \frac{\gamma}{1-\gamma}$.

Proof. (i). Conditional on $\mathbf{X}$: $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}$ (from (40)). By the quadratic form expectation lemma (Lemma 3.7 of the classical notes) with $\mathbb{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$ and $\text{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma_\varepsilon^2 \mathbf{I}_n$:

$$\mathbb{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma_\varepsilon^2 \text{tr}(\mathbf{H}) + \mathbf{0}'\mathbf{H}\mathbf{0} = \sigma_\varepsilon^2 p.$$

Dividing by n gives (i).

(ii). Similarly, $\text{SSE} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$ where $\mathbf{M} = \mathbf{I} - \mathbf{H}$ has $\text{tr}(\mathbf{M}) = n - p$. So $\mathbb{E}[\text{SSE} \mid \mathbf{X}] = \sigma_\varepsilon^2(n - p)$ and $\mathbb{E}[\text{MSE} \mid \mathbf{X}] = \sigma_\varepsilon^2$.

(iii). Since $\mathbf{x}_0$ is independent of $(\mathbf{X}, \boldsymbol{\varepsilon})$ with $\mathbb{E}[\mathbf{x}_0 \mathbf{x}_0'] = \mathbf{I}_p$:

$$\mathbb{E}[(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 \mid \mathbf{X}, \boldsymbol{\varepsilon}] = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})' \mathbb{E}[\mathbf{x}_0 \mathbf{x}_0'] (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2.$$

Using $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$:

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \boldsymbol{\varepsilon}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}'\boldsymbol{\varepsilon}.$$

Let $\mathbf{C} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}'$. By the quadratic form lemma, $\mathbb{E}[\boldsymbol{\varepsilon}'\mathbf{C}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma_\varepsilon^2 \text{tr}(\mathbf{C})$, and using the cyclic property of the trace:

$$\text{tr}(\mathbf{C}) = \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}') = \text{tr}((\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}'\mathbf{X}) = \text{tr}((\mathbf{X}'\mathbf{X})^{-1}).$$

Hence $\mathbb{E}[(\mathbf{x}'_0(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 \text{tr}((\mathbf{X}'\mathbf{X})^{-1})$.

Asymptotic statement. The eigenvalues of $(\mathbf{X}'\mathbf{X})^{-1}$ are the reciprocals of those of $\mathbf{X}'\mathbf{X}$, so

$$\text{tr}((\mathbf{X}'\mathbf{X})^{-1}) = \sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X})} = \frac{1}{n} \sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X}/n)}.$$

We wish to conclude

$$\frac{1}{p} \sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X}/n)} \xrightarrow{\text{a.s.}} \int \frac{1}{x} dF_\gamma(x) = \frac{1}{1-\gamma}, \quad (47)$$

where the moment $\int x^{-1} dF_\gamma(x) = 1/(1-\gamma)$ for $\gamma \in (0, 1)$ is a standard computation via the Stieltjes transform.

Justification of (47). The Marchenko–Pastur law gives weak convergence of the empirical spectral distribution F_p of $\mathbf{X}'\mathbf{X}/n$ to F_γ . Since the test function $g(x) = 1/x$ has a pole at zero, this is *not* immediate from weak convergence and requires uniform truncation of the support away from zero. Under the sub-Gaussian assumption (M2), Theorem 8.2 with $t = O(\sqrt{n})$ gives, for every $\gamma \in (0, 1)$ and all sufficiently large n, p with $p/n \rightarrow \gamma$,

$$\lambda_{\min}(\mathbf{X}'\mathbf{X}/n) \geq (1 - C\sqrt{\gamma})^2 - o(1) =: \underline{\lambda} > 0$$

on an event of probability $1 - 2\exp(-cn)$, which is summable in n ; by Borel–Cantelli, $\lambda_{\min}(\mathbf{X}'\mathbf{X}/n) \geq \underline{\lambda}/2$ for all n large enough almost surely.¹ On this event, all eigenvalues $\lambda_j(\mathbf{X}'\mathbf{X}/n)$ lie in $[\underline{\lambda}/2, M]$ for a deterministic upper bound M (similarly from Theorem 8.2), and the truncated test function $g(x) = 1/x$ is bounded and continuous on this interval. Weak convergence then yields (47).

Combining,

$$\sigma_\varepsilon^2 \text{tr}((\mathbf{X}'\mathbf{X})^{-1}) = \sigma_\varepsilon^2 \cdot \frac{p}{n} \cdot \frac{1}{p} \sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X}/n)} \xrightarrow{\text{a.s.}} \sigma_\varepsilon^2 \cdot \gamma \cdot \frac{1}{1-\gamma} = \sigma_\varepsilon^2 \cdot \frac{\gamma}{1-\gamma}. \quad \square$$

Corollary 15.1 (Estimation Error Diverges as $\gamma \rightarrow 1^-$). Under the conditions of Theorem 15.1, $\mathbb{E}[\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 \text{tr}((\mathbf{X}'\mathbf{X})^{-1}) \rightarrow \infty$ as $p/n \rightarrow 1^-$, since $\frac{1}{1-\gamma} \rightarrow \infty$.

Remark (Diagnosis and cure). Corollary 15.1 is the diagnosis: as the design approaches singularity ($\gamma \rightarrow 1^-$), the smallest eigenvalue of $\mathbf{X}'\mathbf{X}/n$ approaches zero, so $\|(\mathbf{X}'\mathbf{X})^{-1}\|_{\text{op}} \rightarrow \infty$ and OLS estimation error blows up. The standard cure is **ridge regression**: replace $\mathbf{X}'\mathbf{X}$ by $\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}_p$ for some $\lambda > 0$. This shifts every eigenvalue by λ , hence $\|(\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}_p)^{-1}\|_{\text{op}} \leq 1/\lambda$ regardless of how singular $\mathbf{X}'\mathbf{X}$ is. The cost is a bias of order $\lambda \|\boldsymbol{\beta}\|_{\text{op}}$; the optimal λ balances bias against variance and depends on γ , $\|\boldsymbol{\beta}\|_{\text{op}}$, and σ_ε^2 . The bias–variance decomposition is the subject of Exercise 16.11.

Remark (The overparametrized regime $p > n$). When $p > n$, $\mathbf{X}'\mathbf{X}$ is singular and the classical OLS estimator is undefined. The minimum-norm interpolator $\hat{\boldsymbol{\beta}}_{\text{mn}} = \mathbf{X}'(\mathbf{X}\mathbf{X}')^{-1}\mathbf{Y}$ can still be computed, and—surprisingly—its prediction risk can decrease as p/n increases beyond 1. This “double descent” phenomenon is the starting point of modern interpolation theory (Hastie, Montanari, Rosset, and Tibshirani, 2022), but lies beyond the scope of these notes.

16 Exercises

Tier A: Reproduce the Derivation

Exercise 16.1. Let X be a random variable with $\mathbb{E}[X] = 0$ and $a \leq X \leq b$. Prove Hoeffding’s lemma: $\mathbb{E}[e^{tX}] \leq e^{t^2(b-a)^2/8}$.

¹Under the weaker moment assumption of MP (i.i.d. entries with finite fourth moment), the corresponding hard-edge result is the Bai–Yin law (1993). Under (M2), the non-asymptotic bound of Theorem 8.2 suffices.

Exercise 16.2. Let $X_1, \dots, X_n$ be independent with $\mathbb{E}[X_i] = 0$ and $a_i \leq X_i \leq b_i$. Prove Hoeffding's inequality: $\mathbb{P}(|\sum X_i| \geq t) \leq 2 \exp(-2t^2 / \sum (b_i - a_i)^2)$.

Exercise 16.3. Prove that if X is sub-Gaussian, then X^2 is sub-exponential with $\|X^2\|_{\psi_1} \leq C\|X\|_{\psi_2}^2$.

Exercise 16.4. Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be i.i.d. isotropic sub-Gaussian in $\mathbb{R}^p$. Carry out the pointwise Bernstein argument (Step 2 of Theorem 6.1) to show that for a fixed $\mathbf{v} \in S^{p-1}$:

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n [(\mathbf{v}'\mathbf{x}_i)^2 - 1]\right| \geq t\right) \leq 2 \exp\left[-cn \min\left(\frac{t^2}{K^4}, \frac{t}{K^2}\right)\right].$$

Exercise 16.5. Derive the prediction error identity $\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \frac{1}{n}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}$ and compute $\mathbb{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}]$ under sub-Gaussian errors.

Tier B: Apply the Theory

Exercise 16.6. Let $X_1, \dots, X_n$ be i.i.d. Bernoulli(p) with $p \in (0, 1)$. Use Hoeffding's inequality to find the smallest n guaranteeing $\mathbb{P}(|\bar{X} - p| \geq 0.01) \leq 0.05$.

Exercise 16.7. Suppose $\mathbf{x}_i \sim N(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} = \text{diag}(1, 2, 3, \dots, p)$. (a) Compute $\kappa(\boldsymbol{\Sigma})$. (b) How large must n be (in terms of p) to guarantee $\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_{\text{op}}/\|\boldsymbol{\Sigma}\|_{\text{op}} \leq 1/2$ with probability ≥ 0.99 ? (c) Compare the estimation error bound from Theorem 11.1 with the classical variance formula $\sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{11}$.

Exercise 16.8. In the Marchenko–Pastur law with $\gamma = 0.5$, compute $\lambda_- = (1 - \sqrt{0.5})^2$ and $\lambda_+ = (1 + \sqrt{0.5})^2$. What is the asymptotic condition number of $\mathbf{X}'\mathbf{X}/n$? If the true error variance is $\sigma^2 = 1$, estimate $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2$ using Corollary 15.1.

Exercise 16.9. A design matrix $\mathbf{X} \in \mathbb{R}^{100 \times 10}$ has i.i.d. Rademacher entries (± 1 with equal probability). (a) Verify that each row is sub-Gaussian and compute K . (b) Check the sample-size condition $n \geq CK^4p$. (c) Give a high-probability interval for $\lambda_{\min}(\mathbf{X}'\mathbf{X}/n)$ and $\lambda_{\max}(\mathbf{X}'\mathbf{X}/n)$ using Theorem 8.2.

Tier C: Extend and Connect

Exercise 16.10. (Concentration of the MSE.) Under (M1)–(M4), the classical notes show $\mathbb{E}[\text{MSE}] = \sigma^2$ under normality. Use the Hanson–Wright inequality to prove that $|\text{SSE} - \sigma^2(n - p)| \leq C\sigma^2\sqrt{n - p}$ with high probability under sub-Gaussian errors. Conclude that $\text{MSE} = \text{SSE}/(n - p)$ concentrates around σ^2 .

Exercise 16.11. (Ridge regression in the proportional regime.) Consider the ridge estimator $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}'\mathbf{Y}$. (a) Show that $\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}_p)^{-1}(\mathbf{X}'\boldsymbol{\varepsilon} - \lambda\boldsymbol{\beta})$. (b) Decompose $\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2$ into a bias term (depending on $\lambda\boldsymbol{\beta}$) and a variance term (depending on $\boldsymbol{\varepsilon}$). (c) Why does ridge help when $p/n \rightarrow 1^-$?

Exercise 16.12. (Sub-Gaussian tail optimality.) Show that the Hoeffding bound for $\bar{X}$ (Corollary 2.1) cannot be improved beyond constant factors for Rademacher variables: explicitly compute $\mathbb{P}(\sum_{i=1}^n \varepsilon_i \geq t)$ for ε_i i.i.d. uniform on $\{-1, +1\}$ via the MGF, and compare with (6).

Exercise 16.13. (From concentration to confidence intervals.) Under (M1)–(M4), construct a non-asymptotic confidence interval for a single coefficient β_j *without assuming normality*. Specifically: (a) Show $\hat{\beta}_j - \beta_j = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, where $\mathbf{e}_j$ is the j -th standard basis vector. (b) Conditionally on $\mathbf{X}$, bound $|\hat{\beta}_j - \beta_j|$ using the sub-Gaussian tail of $\mathbf{c}'\boldsymbol{\varepsilon}$ for $\mathbf{c} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j$. (c) Show this yields a confidence interval of width $O(\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} \log(1/\alpha)})$ at level $1 - \alpha$. Compare with the classical interval width $t_{\alpha/2, n-p} \cdot s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$.

17 Solutions to Exercises

Solutions to Tier A

Solution to Exercise 16.1. See the proof of Lemma 2.1.

Solution to Exercise 16.2. See the proof of Theorem 2.1.

Solution to Exercise 16.3. See the proof of Proposition 4.1.

Solution to Exercise 16.4. The random variable $Z_i = (\mathbf{v}'\mathbf{x}_i)^2 - 1$ satisfies $\mathbb{E}[Z_i] = 0$ (isotropic assumption) and is sub-exponential with $\|Z_i\|_{\psi_1} \leq CK^2$ (since $\mathbf{v}'\mathbf{x}_i$ is sub-Gaussian with norm $\leq K$ and squares of sub-Gaussians are sub-exponential by Proposition 4.1). By Bernstein's inequality (Theorem 4.1) applied to $\sum_{i=1}^n Z_i$:

$$\mathbb{P}\left(\left|\sum_{i=1}^n Z_i\right| \geq nt\right) \leq 2 \exp\left[-c \min\left(\frac{n^2 t^2}{n C^2 K^4}, \frac{nt}{C K^2}\right)\right] = 2 \exp\left[-cn \min\left(\frac{t^2}{K^4}, \frac{t}{K^2}\right)\right].$$

Solution to Exercise 16.5. Substituting $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$:

$$\mathbf{X}(\hat{\beta} - \beta) = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} = \mathbf{H}\boldsymbol{\varepsilon}.$$

Hence $\frac{1}{n}\|\mathbf{X}(\hat{\beta} - \beta)\|^2 = \frac{1}{n}\|\mathbf{H}\boldsymbol{\varepsilon}\|^2 = \frac{1}{n}(\mathbf{H}\boldsymbol{\varepsilon})'\mathbf{H}\boldsymbol{\varepsilon} = \frac{1}{n}\boldsymbol{\varepsilon}'\mathbf{H}'\mathbf{H}\boldsymbol{\varepsilon} = \frac{1}{n}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}$ (using $\mathbf{H}' = \mathbf{H}$ and $\mathbf{H}^2 = \mathbf{H}$). For the conditional expectation: since $\varepsilon_1, \dots, \varepsilon_n$ are independent with $\mathbb{E}[\varepsilon_i] = 0$ and $\mathbb{E}[\varepsilon_i^2] = \sigma_\varepsilon^2$, we have $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}] = \sigma_\varepsilon^2 \mathbf{I}_n$. By Lemma 3.7 of the classical notes (expected value of quadratic form, with $\boldsymbol{\mu} = \mathbf{0}$):

$$\mathbb{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} | \mathbf{X}] = \text{tr}(\mathbf{H} \cdot \sigma_\varepsilon^2 \mathbf{I}_n) = \sigma_\varepsilon^2 \text{tr}(\mathbf{H}) = \sigma_\varepsilon^2 p.$$

Dividing by n : $\mathbb{E}[\frac{1}{n}\|\mathbf{X}(\hat{\beta} - \beta)\|^2 | \mathbf{X}] = \sigma_\varepsilon^2 p/n$.

Solutions to Tier B

Solution to Exercise 16.6. Each $X_i \in [0, 1]$, so $b_i - a_i = 1$. By Corollary 2.1: $\mathbb{P}(|\bar{X} - p| \geq 0.01) \leq 2 \exp(-2n(0.01)^2) = 2e^{-n/5000}$. Setting $2e^{-n/5000} \leq 0.05$ gives $n \geq 5000 \ln(40) \approx 18444$.

Solution to Exercise 16.7. (a) $\kappa(\boldsymbol{\Sigma}) = \lambda_{\max}/\lambda_{\min} = p/1 = p$. (b) The whitened vectors $\mathbf{z}_i = \boldsymbol{\Sigma}^{-1/2}\mathbf{x}_i$ are isotropic Gaussian with $\|\mathbf{z}_i\|_{\psi_2} \leq C$. The condition $\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_{\text{op}}/\|\boldsymbol{\Sigma}\|_{\text{op}} \leq 1/2$ reduces (after whitening) to $\|\hat{\mathbf{I}} - \mathbf{I}_p\|_{\text{op}} \leq 1/2$ for the whitened sample covariance. By Theorem 6.1 with $\delta = 1/2$, we need $n \geq 4Cp$. For probability 0.99, we additionally need $2 \exp(-cp) \leq 0.01$, i.e., $p \geq \frac{1}{c} \ln 200$, which is mild.

(c) The classical formula gives $\text{var}(\hat{\beta}_1) = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{11}$, which depends on the specific realization of $\mathbf{X}$. For the diagonal covariance, $\mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}] \approx \frac{1}{n}\boldsymbol{\Sigma}^{-1}$ for large n , so $\text{var}(\hat{\beta}_1) \approx \sigma^2/(n \cdot 1) = \sigma^2/n$. The modern bound gives $\|\hat{\beta} - \beta\|^2 \leq C\sigma^2 p/(n\lambda_{\min}(\boldsymbol{\Sigma})) = C\sigma^2 p/n$, which is larger by a factor p (but bounds the total ℓ^2 error, not just one coordinate).

Solution to Exercise 16.8. $\lambda_- = (1 - 1/\sqrt{2})^2 = (2 - \sqrt{2})^2/4 \approx 0.0858$. $\lambda_+ = (1 + 1/\sqrt{2})^2 = (2 + \sqrt{2})^2/4 \approx 2.914$. Asymptotic condition number: $\kappa \approx \lambda_+/\lambda_- \approx 33.97$. For the estimation error in the isotropic case ($\boldsymbol{\Sigma} = \mathbf{I}_p$, $\sigma^2 = 1$), Theorem 15.1(iii) gives, as $p/n \rightarrow \gamma = 0.5$,

$$\mathbb{E}[\|\hat{\beta} - \beta\|^2 | \mathbf{X}] \rightarrow \sigma_\varepsilon^2 \cdot \frac{\gamma}{1 - \gamma} = 1 \cdot \frac{0.5}{0.5} = 1.$$

Solution to Exercise 16.9. (a) Each entry X_{ij} is Rademacher: $\|X_{ij}\|_{\psi_2} = C$ (by Example 1.1(2)). A row $\mathbf{x}_i = (X_{i1}, \dots, X_{ip})'$ has $\|\langle \mathbf{x}_i, \mathbf{v} \rangle\|_{\psi_2} = \|\sum_j v_j X_{ij}\|_{\psi_2} \leq C\|\mathbf{v}\|$ by Proposition 1.1. So $K = C$. (b) $n = 100$, $p = 10$, $K^4 p = C^4 \cdot 10$. For reasonable C , $n = 100 \gg C^4 \cdot 10$, so the condition is satisfied. (c) By (36) with $t = 0$: $\lambda_{\min}(\mathbf{X}'\mathbf{X}/n) \gtrsim (1 - C'\sqrt{10/100})^2 = (1 - C'\sqrt{0.1})^2$ and $\lambda_{\max}(\mathbf{X}'\mathbf{X}/n) \lesssim (1 + C'\sqrt{0.1})^2$. With $C' \approx 1$: $\lambda_{\min} \gtrsim (1 - 0.316)^2 \approx 0.467$, $\lambda_{\max} \lesssim (1 + 0.316)^2 \approx 1.732$.

Solutions to Tier C

Solution to Exercise 16.10. We have $\text{SSE} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$ where $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$ is symmetric idempotent with $\text{tr}(\mathbf{M}) = n - p$ and $\|\mathbf{M}\|_{\text{op}} = 1$, $\|\mathbf{M}\|_{\text{F}}^2 = n - p$. By Hanson–Wright (Theorem 3.3), conditionally on $\mathbf{X}$:

$$\mathbb{P}(|\text{SSE} - \sigma_\varepsilon^2(n - p)| \geq t \mid \mathbf{X}) \leq 2 \exp \left[-c \min \left(\frac{t^2}{\sigma_\varepsilon^4(n - p)}, \frac{t}{\sigma_\varepsilon^2} \right) \right].$$

Setting $t = \delta \sigma_\varepsilon^2(n - p)$: $\mathbb{P}(|\text{SSE} - \sigma_\varepsilon^2(n - p)| \geq \delta \sigma_\varepsilon^2(n - p)) \leq 2 \exp(-c \min(\delta^2(n - p), \delta(n - p))) \leq 2 \exp(-c' \delta^2(n - p))$. With $\delta = C/\sqrt{n - p}$: $|\text{SSE} - \sigma_\varepsilon^2(n - p)| \leq C \sigma_\varepsilon^2 \sqrt{n - p}$ with probability $\geq 1 - 2e^{-c'C^2}$. Hence $|\text{MSE} - \sigma_\varepsilon^2| = |\text{SSE}/(n - p) - \sigma_\varepsilon^2| \leq C \sigma_\varepsilon^2 / \sqrt{n - p} \rightarrow 0$.

Solution to Exercise 16.11.

Part (a): the residual identity. Starting from $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}'\mathbf{Y} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})$, write $\mathbf{X}'\mathbf{X} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}) - \lambda \mathbf{I}$. Then

$$(\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta} - \lambda(\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \boldsymbol{\beta},$$

so $\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}'\boldsymbol{\varepsilon} - \lambda(\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \boldsymbol{\beta}$.

Part (b): bias–variance decomposition. Set $\mathbf{R}_\lambda := (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1}$. Then $\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} = \mathbf{R}_\lambda \mathbf{X}'\boldsymbol{\varepsilon} - \lambda \mathbf{R}_\lambda \boldsymbol{\beta}$, and

$$\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 = \lambda^2 \boldsymbol{\beta}' \mathbf{R}_\lambda^2 \boldsymbol{\beta} - 2\lambda \boldsymbol{\beta}' \mathbf{R}_\lambda^2 \mathbf{X}'\boldsymbol{\varepsilon} + \boldsymbol{\varepsilon}' \mathbf{X} \mathbf{R}_\lambda^2 \mathbf{X}' \boldsymbol{\varepsilon}.$$

Taking conditional expectation given $\mathbf{X}$: the cross term vanishes since $\mathbb{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$, so

$$\mathbb{E}[\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 \mid \mathbf{X}] = \underbrace{\lambda^2 \boldsymbol{\beta}' \mathbf{R}_\lambda^2 \boldsymbol{\beta}}_{\text{bias}^2(\lambda)} + \underbrace{\sigma_\varepsilon^2 \text{tr}(\mathbf{X} \mathbf{R}_\lambda^2 \mathbf{X}')}_{\text{variance}(\lambda)}, \quad (48)$$

where the second equality uses $\mathbb{E}[\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}' \mid \mathbf{X}] = \sigma_\varepsilon^2 \mathbf{I}_n$ and the cyclic property of the trace.

Spectral form. Let $\mathbf{X}'\mathbf{X} = \mathbf{U} \text{diag}(\mu_1, \dots, \mu_p) \mathbf{U}'$ be the spectral decomposition, and write $\tilde{\boldsymbol{\beta}} := \mathbf{U}'\boldsymbol{\beta}$ in the eigenbasis. Then $\mathbf{R}_\lambda = \mathbf{U} \text{diag}(1/(\mu_j + \lambda)) \mathbf{U}'$, and:

$$\text{bias}^2(\lambda) = \lambda^2 \sum_{j=1}^p \frac{\tilde{b}_j^2}{(\mu_j + \lambda)^2}, \quad (49)$$

$$\text{variance}(\lambda) = \sigma_\varepsilon^2 \sum_{j=1}^p \frac{\mu_j}{(\mu_j + \lambda)^2}, \quad (50)$$

the last using $\text{tr}(\mathbf{X} \mathbf{R}_\lambda^2 \mathbf{X}') = \text{tr}(\mathbf{R}_\lambda^2 \mathbf{X}'\mathbf{X}) = \sum_j \mu_j / (\mu_j + \lambda)^2$.

Two uniform bounds. For each j , the function $\lambda \mapsto \lambda^2 / (\mu_j + \lambda)^2$ is monotone increasing in λ and bounded by 1, so

$$\text{bias}^2(\lambda) \leq \|\boldsymbol{\beta}\|_{\text{op}}^2 \quad \text{for every } \lambda \geq 0. \quad (51)$$

For each j , the function $\mu \mapsto \mu / (\mu + \lambda)^2$ is maximized at $\mu = \lambda$ with maximum value $1/(4\lambda)$, so $\mu_j / (\mu_j + \lambda)^2 \leq 1/(4\lambda)$, giving

$$\text{variance}(\lambda) \leq \frac{\sigma_\varepsilon^2 p}{4\lambda} \quad \text{for every } \lambda > 0. \quad (52)$$

The crucial contrast with OLS. Bounds (51)–(52) do not depend on $\lambda_{\min}(\mathbf{X}'\mathbf{X})$. Combining,

$$\mathbb{E}[\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 \mid \mathbf{X}] \leq \|\boldsymbol{\beta}\|_{\text{op}}^2 + \frac{\sigma_\varepsilon^2 p}{4\lambda}$$

for every $\lambda > 0$, regardless of how singular $\mathbf{X}'\mathbf{X}$ is. Setting $\lambda \rightarrow \infty$ kills the variance; setting $\lambda \rightarrow 0^+$ recovers the unbounded OLS variance (cf. Corollary 15.1). In contrast, OLS has $\mathbb{E}[\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 \text{tr}((\mathbf{X}'\mathbf{X})^{-1}) = \sigma_\varepsilon^2 \sum_j 1/\mu_j$, which diverges as $\mu_p = \lambda_{\min}(\mathbf{X}'\mathbf{X}) \rightarrow 0$.

Part (c): why ridge helps as $\gamma \rightarrow 1^-$. In OLS, the variance is dominated by the smallest eigenvalue μ_p : as $\gamma \rightarrow 1^-$, $\mu_p/n \rightarrow (1 - \sqrt{\gamma})^2 \rightarrow 0$ and the variance term $1/\mu_p$ explodes. Ridge regularization replaces $1/\mu_j$ by $\mu_j/(\mu_j + \lambda)^2$ in each spectral component, which is bounded by $1/(4\lambda)$ regardless of μ_j . The cost is a bias term, but (51) shows the bias is universally bounded by $\|\boldsymbol{\beta}\|_{\text{op}}^2$. Choosing λ to balance the two leads to a finite-risk estimator even at $\gamma = 1$.

Optimal λ (sketch). Differentiating $\|\boldsymbol{\beta}\|_{\text{op}}^2 + \sigma_\varepsilon^2 p/(4\lambda)$ in λ does not yield a finite minimizer (the bound is monotone decreasing in λ within $\lambda > 0$, but with the actual bias (49) the trade-off is delicate). Under the isotropic-design assumption $\mathbf{X}'\mathbf{X}/n \approx \mathbf{I}_p$ (so $\mu_j \approx n$ for all j), the bias (49) simplifies to $\lambda^2 \|\boldsymbol{\beta}\|_{\text{op}}^2/(n + \lambda)^2$ and the variance to $\sigma_\varepsilon^2 p \cdot n/(n + \lambda)^2$; the sum is minimized at $\lambda^* = p\sigma_\varepsilon^2/\|\boldsymbol{\beta}\|_{\text{op}}^2$, giving total risk $\asymp \|\boldsymbol{\beta}\|_{\text{op}}^2 \sigma_\varepsilon^2 p/(n\|\boldsymbol{\beta}\|_{\text{op}}^2 + \sigma_\varepsilon^2 p)$. This is bounded uniformly in γ .

Solution to Exercise 16.12. For Rademacher ε_i , $S_n = \sum \varepsilon_i$. The MGF is $\mathbb{E}[e^{tS_n}] = (\cosh t)^n$. By the Chernoff bound: $\mathbb{P}(S_n \geq s) \leq \inf_{t>0} e^{-ts} (\cosh t)^n$. Using $\cosh t \leq e^{t^2/2}$ gives $\mathbb{P}(S_n \geq s) \leq e^{-s^2/(2n)}$, matching Hoeffding. The bound $\cosh t \leq e^{t^2/2}$ is tight at $t = 0$ (both sides equal 1, derivatives match), so the Hoeffding bound is essentially optimal for Rademacher sums.

Solution to Exercise 16.13. (a) $\hat{\beta}_j = \mathbf{e}_j' \hat{\boldsymbol{\beta}} = \mathbf{e}_j' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y} = \beta_j + \mathbf{e}_j' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\boldsymbol{\varepsilon} = \beta_j + \mathbf{c}'\boldsymbol{\varepsilon}$ where $\mathbf{c} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{e}_j$.

(b) Conditionally on $\mathbf{X}$, $\mathbf{c}'\boldsymbol{\varepsilon} = \sum_i c_i \varepsilon_i$ is a sum of independent sub-Gaussian variables with $\|\mathbf{c}'\boldsymbol{\varepsilon}\|_{\psi_2} \leq C\sigma\|\mathbf{c}\|$. By the sub-Gaussian tail bound:

$$\mathbb{P}(|\hat{\beta}_j - \beta_j| \geq t \mid \mathbf{X}) \leq 2 \exp\left(-\frac{ct^2}{\sigma^2 \|\mathbf{c}\|^2}\right).$$

Now $\|\mathbf{c}\|^2 = \mathbf{c}'\mathbf{c} = \mathbf{e}_j' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} \mathbf{e}_j = [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$.

(c) Setting $2 \exp(-ct^2/(\sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}]_{jj})) = \alpha$ gives $t = \sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} \cdot \frac{1}{c} \ln(2/\alpha)}$. The confidence interval is:

$$\hat{\beta}_j \pm \sigma \sqrt{\frac{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}{c} \ln \frac{2}{\alpha}}.$$

Comparison. The classical interval has width $\propto t_{\alpha/2, n-p} \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$, using s in place of σ . For large $n - p$, $t_{\alpha/2, n-p} \approx z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2) \approx \sqrt{2 \ln(2/\alpha)}$ (by the Gaussian tail approximation). So the modern and classical widths agree up to constants. The modern bound has two advantages: (1) it does not require normality, and (2) it is a finite-sample statement, not asymptotic. Its disadvantage: it uses the unknown σ rather than the data-driven s (though s concentrates around σ by Exercise 16.10).